WFP
World Food Programme

SAVING LIVES
CHANGING LIVES

Early Warning and Forecasting models of Food Insecurity

2026 April

Our Expertise & Team

An inter-disciplinary team bridging domain expertise with data science and software development – to transform ideas into end-to-end products.

Kyriacos
Head of Unit

ANALYSTS

Marie
Lead

Anja
Conflict & displacement

Amanda
Economics

Diego
Natural hazards

DATA SCIENTISTS

Duccio
Lead

Alice
Food security forecasting

Vasiliki
Nutrition forecasting

Eirik
Nutrition forecasting

Jonas
Food Security Modeling

Diego
Food Security Modeling

SOFTWARE DEVELOPERS

Kingston
Lead

Franklin
Frontend development

PROJECT MANAGEMENT & PARTNERSHIPS

Sandra
Project management

Chiara
Partnerships

Key Target Indicators of Food Insecurity

INDICATOR CATEGORY	KEY MEASURES	TYPICAL TARGET THRESHOLDS	PRIMARY USE
IPC / CH Classification	IPC or CH Phases	Phase 3 (Crisis) and above	Strategic targeting, prioritization, response planning
Food Consumption	Food Consumption Score (FCS)	Poor and Borderline FCS	Household food access and diet quality assessment
Coping Strategies	Reduced CSI, Livelihood CSI	High or worsening scores	Stress and crisis behavior monitoring, resilience analysis
Nutrition Status	Micronutrient Inadequacy, GAM and SAM	GAM ≥ 10–15% or MUAC <125 mm	Identify acute malnutrition and health risk

Key Inputs

Humanitarian Assistance

- In Kind Assistance
- CBT Assistance

1

WFP

Food prices and Affordability

- Global Food and Fuel Prices
- Local Market Prices

2

WFP / WB/FAO

Climate Conditions

- SPI
- NDVI
- Flood triggers
- Livelihood Zones

3

CHIRPS and MODIS

IDPs and Returnees

- IDP count per LGA
- Returnee count

4

IOM

Conflict

- Fatalities
- Conflict Event
- Battles
- Violence against Civilians

5

ACLED / UCPD

Modelling Approaches for Early Warning Signals

Anomaly Detection

Trend or seasonality breaks

Change-point detection

Isolation Forest / Autoencoders

Monitoring & Alerting

- Threshold-based rules
- Z-score / EWMA monitoring

Point Forecasting

- ARIMA / SARIMA
- Tree-based models (RF, GBM)
- Short-tarm outlooks

Probabilistic Modelling

- Bayesian regression
- Quantile forecasts
- Uncertainty / risk bands

Hybrid/Ensemble Approaches

- Multi-model ensembles
- Scenario-based signals
- Human-in-the-loop validation

Identified Best Practices

- **Start from decisions, not models**: Clearly define the operational or strategic decisions the early-warning system is meant to inform, and design models to support those decisions rather than prediction for its own sake.

- **Embed domain expertise from the outset**: Food security and nutrition experts must be involved from problem formulation through interpretation, ensuring models reflect livelihoods, seasonality, shocks, and operational realities.

- **Explicitly map needs to decision pathways**: Identify the specific needs being targeted and the decision-making processes to be influenced, including who acts, at what geographic level, and on what time horizon.

- **Be realistic about data constraints**: Align modelling ambition with what available data can credibly support in terms of quality, coverage, frequency of updates, and timeliness, prioritizing robustness over false precision.

- **Use models to structure risk, not replace judgement**: Modelling outputs should support transparent, risk-based discussions and human-in-the-loop decisions, rather than automate responses in complex food security contexts.

Yemen Poor and Borderline Food Consumption Forecast

Lead Time: Quarter/Semester
Granularity: adm2

FCS forecasting > Yemen - admin2 – Quarterly

Model overview

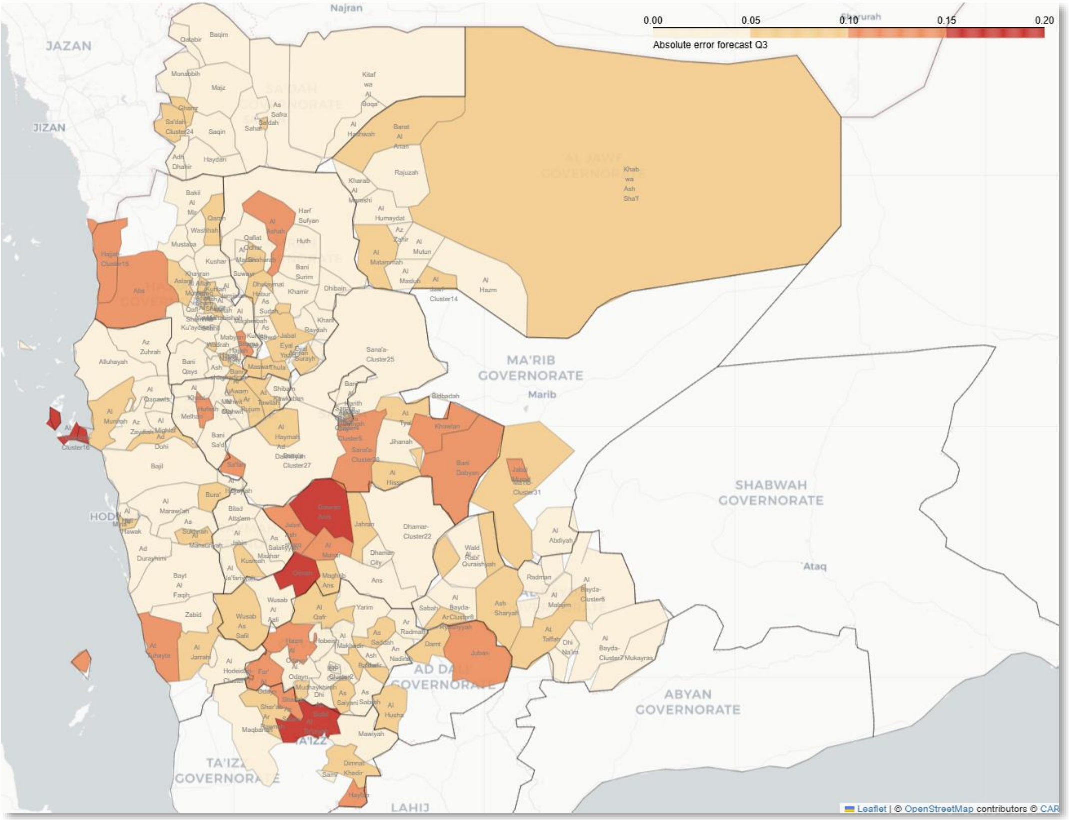
- Quarterly forecasts of population in poor or borderline food consumption (FCS) for every admin2 in Yemen North and South.
- Among the variables used, HFA stands out as the model has learned that its presence reduces the level of food insecurity.

Performance and validation

- Validation on completed quarters shows an average absolute error of 4.8%, indicating solid reliability.
- Two iterations completed, integrating country office feedback and context-specific variables
- Model was validated against the completed quarter and IPC-projected food security levels, and feedback will guide next iterations.

Next Steps

- Model FCG deteriorations (e.g., FCG 2 → FCG 1) developing an IPC-compatible model



Example of Q3 2025 Validation
Overall low error: The map shows that in most districts, the difference between the forecast and the actual value is small (lighter colors).

How we predict

FCG<=2 at 2026Q3 <- (model) **features at 2026Q1** + **features at 2026Q3**

features at 2026Q1:

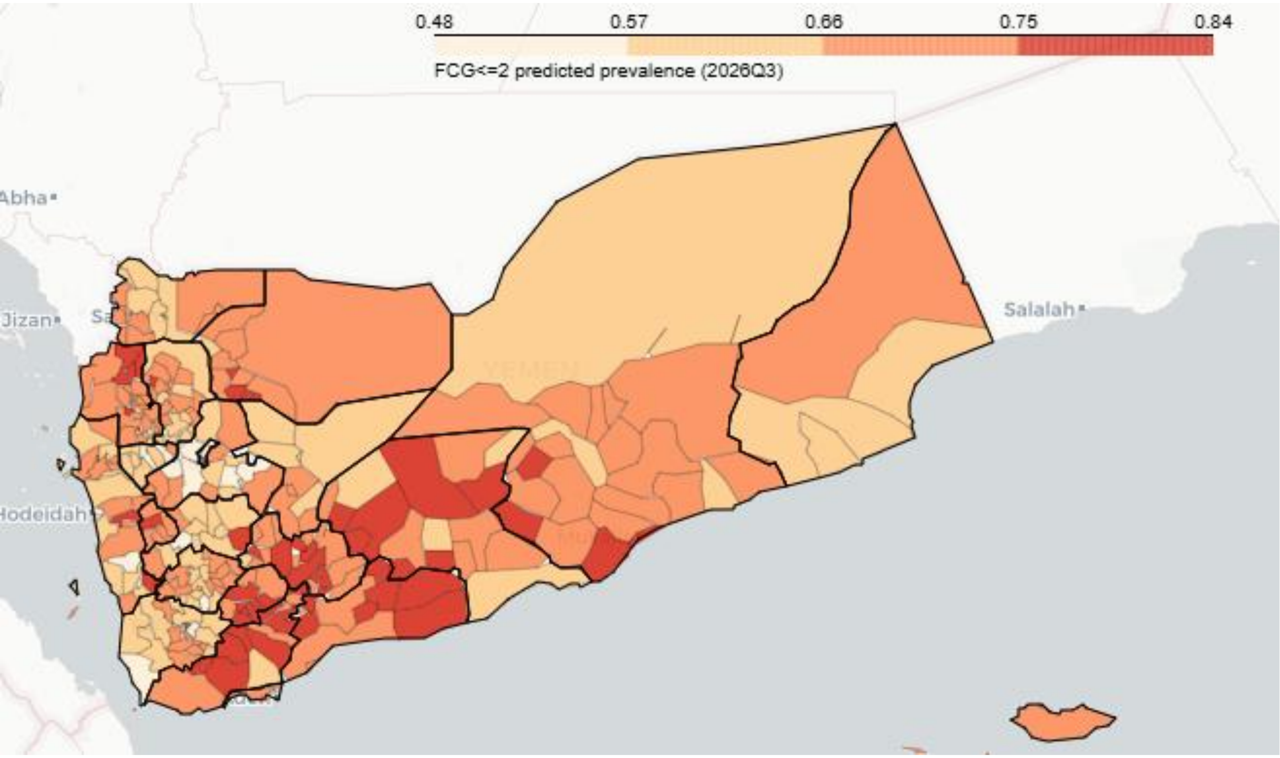
FCG<=2
rCSI
Prices
Conflict
Displacement
Climate
…

features at 2026Q3:

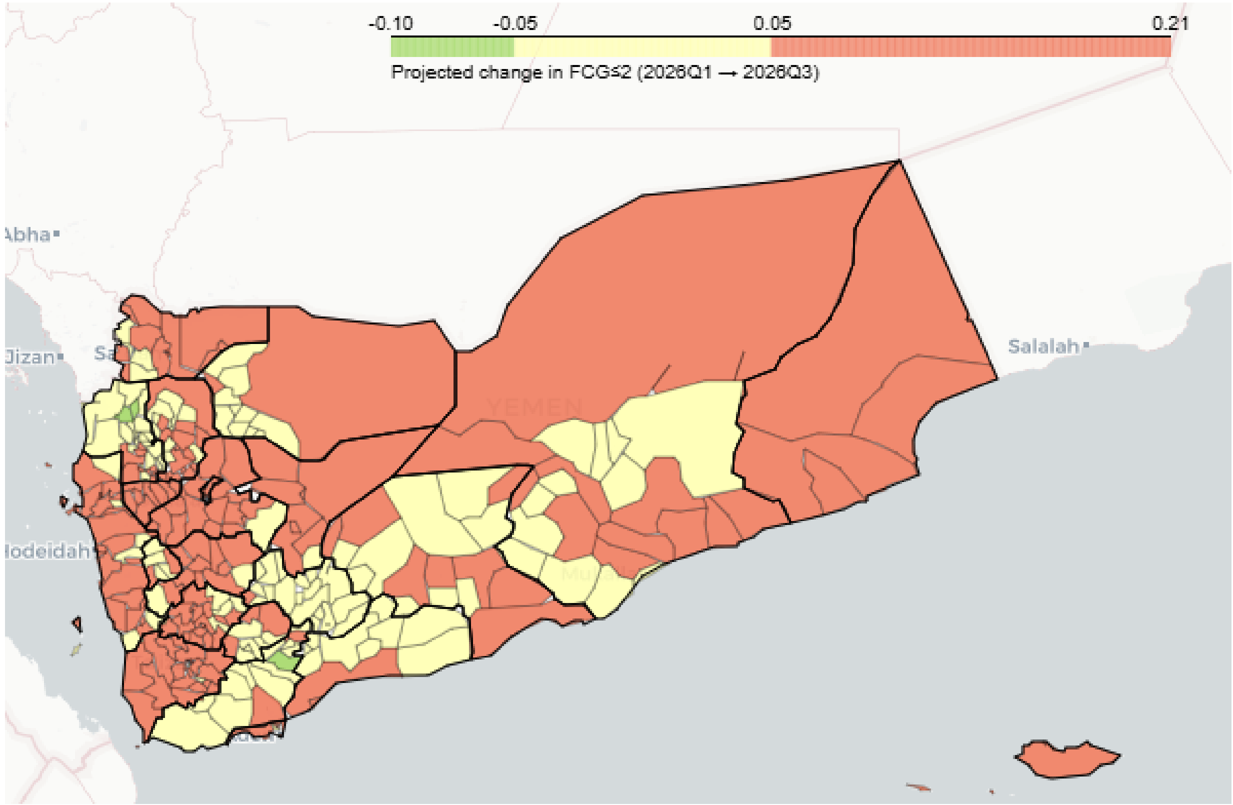
Planned assistance
Ramadan

Q3-2026 forecast by District

Inadequate: FCG<=2



Delta (Q1 -> Q3)



communications earth & environment

Article

https://doi.org/10.1038/s43247-024-01698-9

Forecasting trends in food security with real time data

Check for updates

Joschka Herteux[1], Christoph Raeth[2], Giulia Martini[1], Amine Baha[3], Kyriacos Kouppari[1], Ilaria Lauzana[1] & Duccio Piovani[1]

Early warning systems are an essential tool for effective humanitarian action. Advance warnings on impending disasters facilitate timely and targeted response which help save lives and livelihoods. In this work we present a quantitative methodology to forecast levels of food consumption for 60 consecutive days, at the sub-national level, in four countries: Mali, Nigeria, Syria, and Yemen. The methodology is built on publicly available data from the World Food Programme's global hunger monitoring system which collects, processes, and displays daily updates on key food security metrics, conflict, weather events, and other drivers of food insecurity. In this study we assessed the performance of various models including Autoregressive Integrated Moving Average (ARIMA), Extreme Gradient Boosting (XGBoost), Long Short Term Memory (LSTM) Network, Convolutional Neural Network (CNN), and Reservoir Computing (RC), by comparing their Root Mean Squared Error (RMSE) metrics. Our findings highlight Reservoir Computing as a particularly well-suited model in the field of food security given both its notable resistance to over-fitting on limited data samples and its efficient training capabilities. The methodology we introduce establishes the groundwork for a global, data-driven early warning system designed to anticipate and detect food insecurity.

WFP
World Food Programme

SAVING LIVES
CHANGING LIVES

IRG Governorates

Governorate	Direction	Δ FCG≤2	# Clust.	Feature 1	Feature 2	Feature 3
Al Maharah	Worsening	+8.9%	1	↑ FCG Baseline	↑ Global Fuel Prices (max.)	↑ Seasonality
Taizz	Worsening	+8.8%	21	↑ FCG Baseline	↑ Global Fuel Prices (max.)	↑ Seasonality
Marib	Worsening	+7.4%	6	↑ FCG Baseline	↑ Global Fuel Prices (max.)	↑ Seasonality
Aden	Worsening	+7.2%	8	↑ FCG Baseline	↑ Global Fuel Prices (max.)	↑ Seasonality
Shabwah	Stable	+4.6%	17	↑ FCG Baseline	↑ Global Fuel Prices (max.)	↑ Seasonality
Hadramaut	Stable	+4.5%	21	↓ FCG Baseline	↑ Global Fuel Prices (max.)	↑ Seasonality
Al Dhale'e	Stable	+3.2%	9	↓ FCG Baseline	↑ Global Fuel Prices (max.)	↑ Seasonality
Abyan	Stable	+3.1%	11	↓ FCG Baseline	↑ Global Fuel Prices (max.)	↑ Seasonality
Lahj	Stable	+2.5%	15	↓ FCG Baseline	↑ Global Fuel Prices (max.)	↑ Seasonality

1 cluster (Habil Jabr) is expected to improve due to very high (91%) levels of insufficient food consumption in 2026Q1

IRG Overview

54 clusters deteriorating · **51 clusters** stable · **1 cluster** improving · Largest deteriorations: **Al Maharah +8.9pp**, **Taizz +8.8pp**, **Marib +7.4pp**

01 Baseline Insufficient Food Consumption

98/106 clusters

.

02 Global Fuel Prices (max.)

102/106 clusters

Significantly higher maximum fuel prices in 2026Q1 are expected to lead to increased levels of insufficient food consumption.

03 Seasonality

105/106 clusters

Seasonal deteriorations expected in line with the lean season. Deterioration in Q# compared to the current Q1 improved levels of insufficient food consumption influenced by ramadan.

04 r3-month rainfall anomalies

~58 clusters · mixed direction

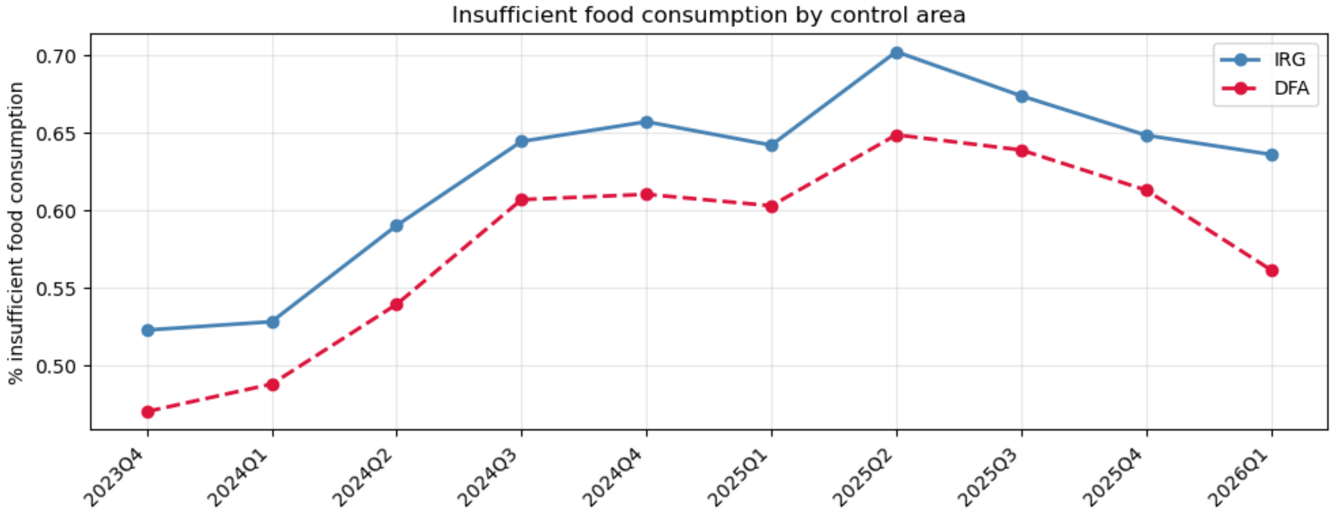
3-month rainfall anomalies have different impacts across the clusters. Certain clusters experienced favourable rainfall, while others did not. Explains the stabilising climate signal in Aden and Taizz clusters.

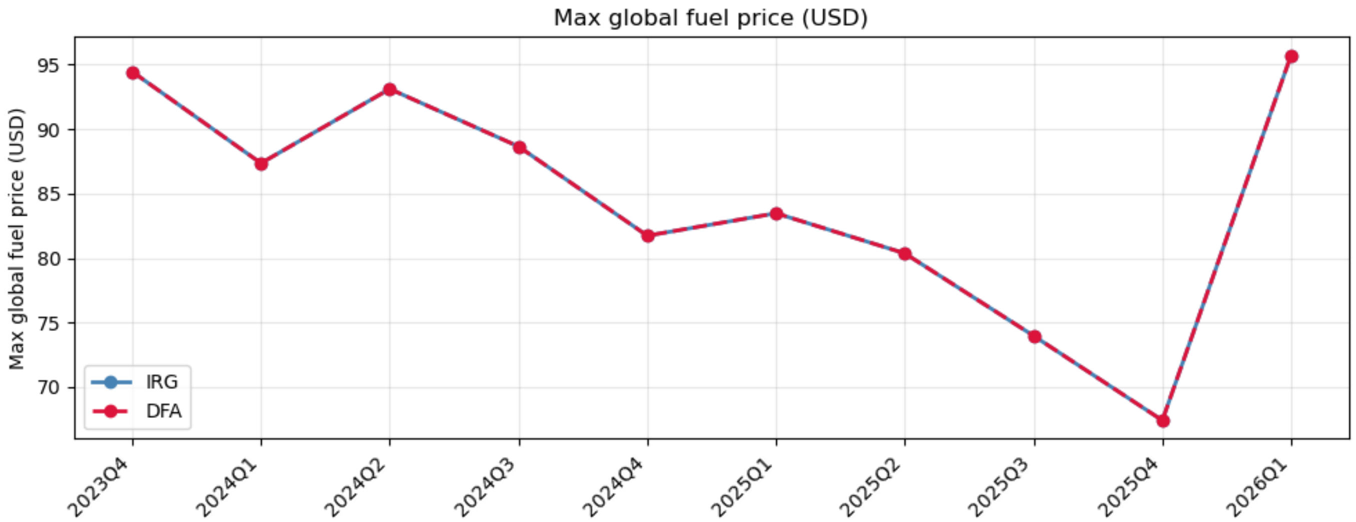
05 Local fuel prices

~20 clusters · mostly worsening

Local fuel price and affordability measures, complementing global_fuel_max in clusters with more localised price dynamics, particularly in Abyan and Taizz governorates.

Conflict and displacement show no predictive signal in this forecast round. · 'Active' = feature/driver ranks among the three largest SHAP contributors for that cluster.





Main Drivers of Deterioration

Seasonality

Levels of insufficient food consumption have significantly come down in 2026Q1 in line with the effects of the post-harvest season and the ramadan period.

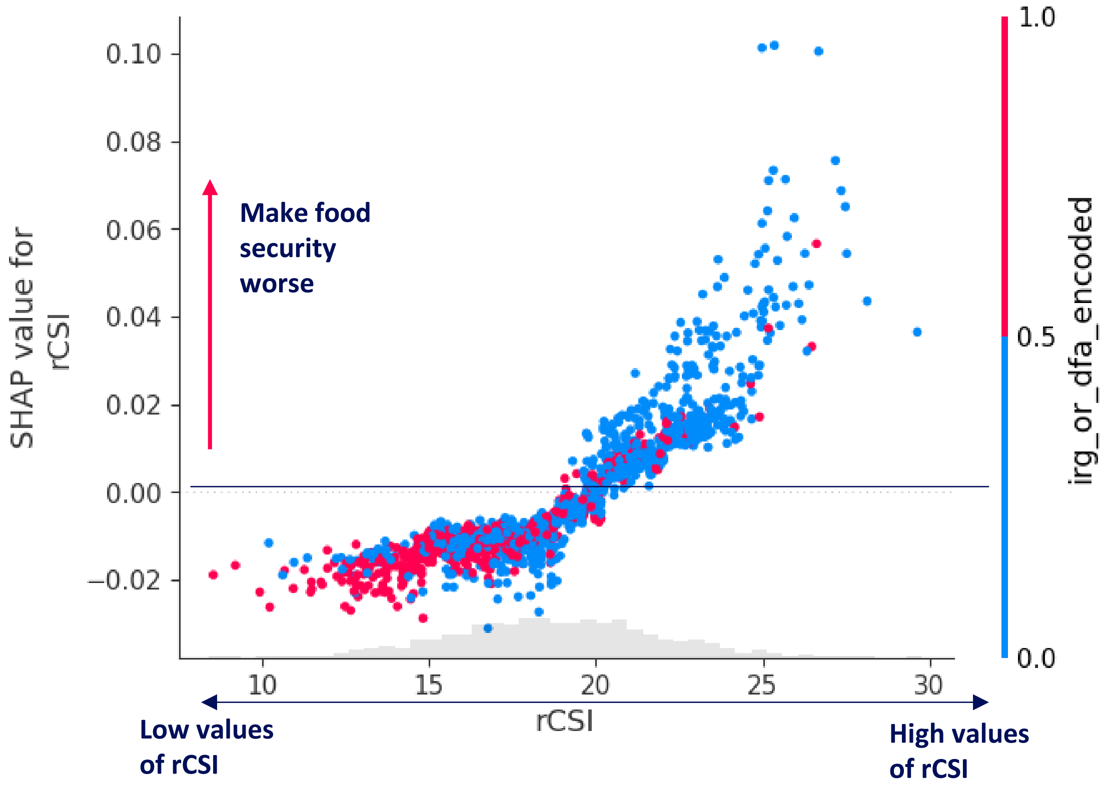
In 2026Q3 insufficient food consumption is expected to increase again in line with the lean season, and al ack of the positive effects experienced during the ramadan period.

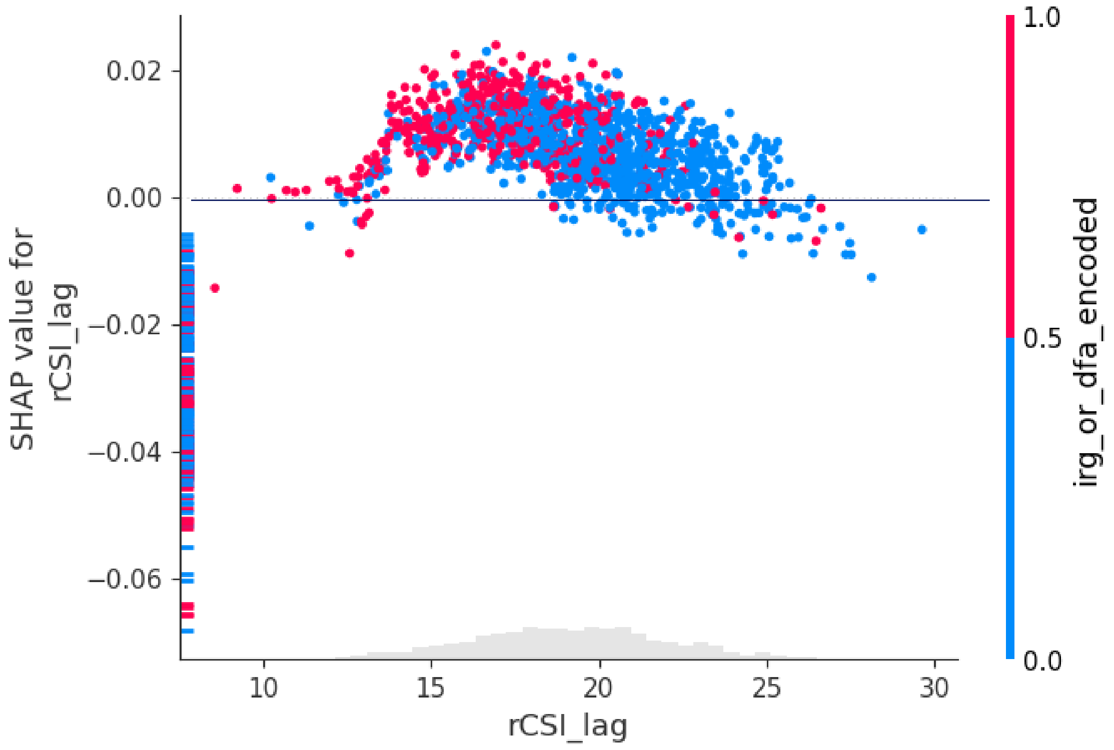
Global Fuel Prices

The conflict in the middle east has led to sudden global fuel price increases. These price increases, if consistent, are expected to drive higher levels of insufficient food consumption by 2026Q3

Model learnings: rCSI

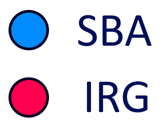
What the model has learned about how drivers influence the outcome variable





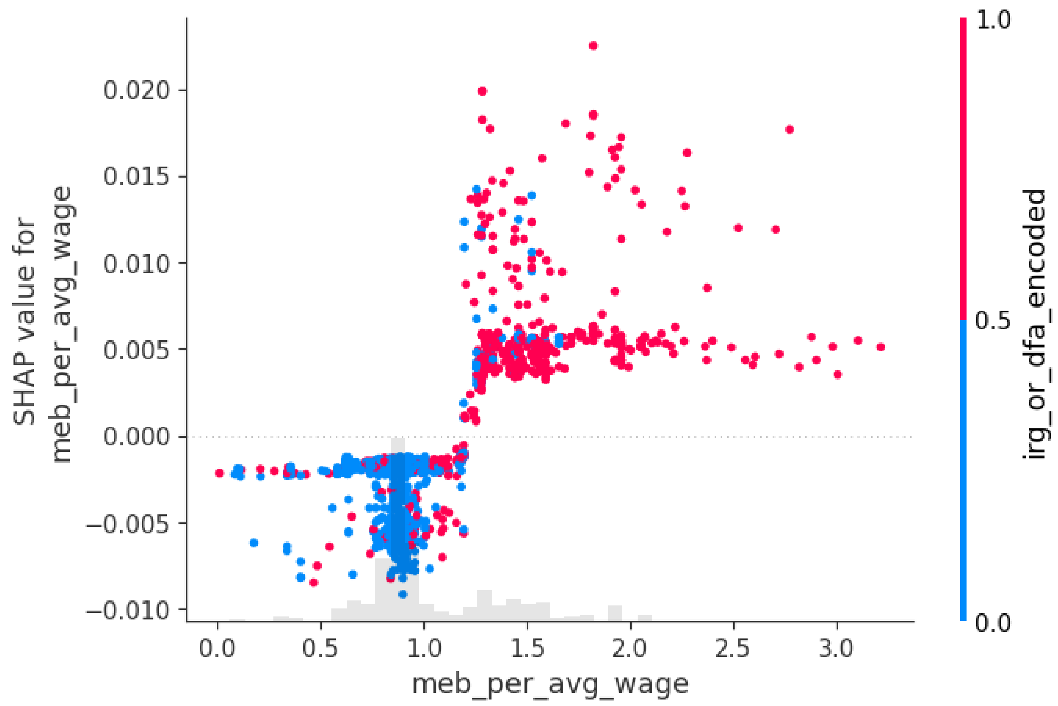
Current rCSI reflects immediate household stress: low levels indicate better food security, while high levels signal acute pressure as families reduce meals or adopt negative coping behaviors.

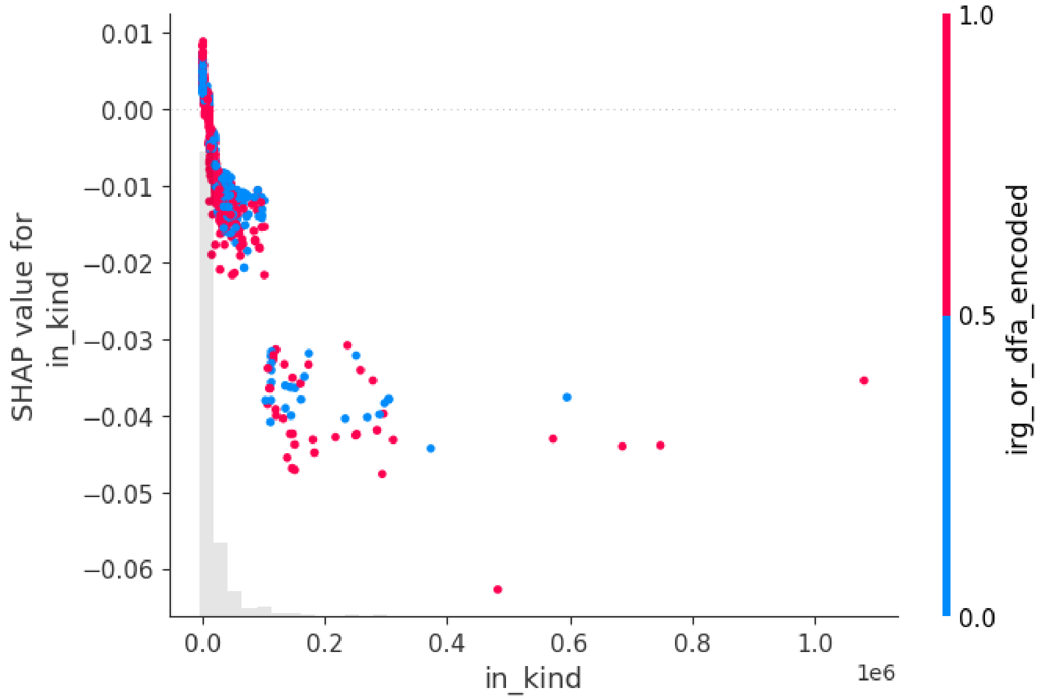
Lagged rCSI captures persistent vulnerability: when coping strategies persist across quarters, the model interprets this as stress that has already materialised, worsening current food security. The model has learned that SBA districts systematically rely on higher coping levels



Model learnings: affordability and aid

What the model has learned about how drivers influence the outcome variable





The model captures affordability by comparing the minimum food basket to average wages, learning that lower affordability values signal stronger economic pressure and are consistently associated with worse food security outcomes.

A value of 1 means a household needs 100% of its salary to afford the MEB.

The feature represents the number of beneficiaries receiving in-kind assistance, with the model learning that moderate levels tend to improve food security, while very high values may reflect crisis response and correlate with entrenched vulnerability.

Nigeria CH Phases Monitoring

Lead Time: Nowcasting
Granularity: adm2

IPC/CH nowcasting > Nigeria – admin2

Model Overview

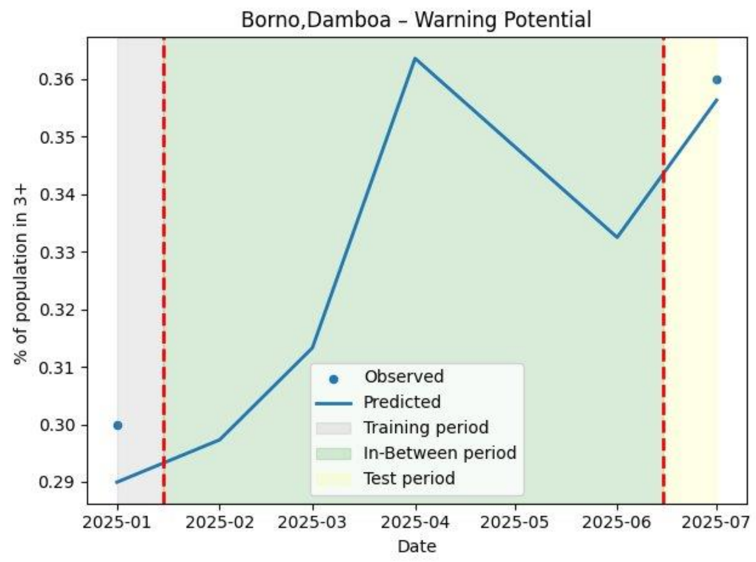
- Uses real-time and secondary indicators to fill data gaps and update CH phases at admin2 level, in the north of Nigeria.
- Supports decision-making by filling information gaps between CH assessment periods.
- Allows country office to act proactively and plan interventions based on evolving conditions

Approach

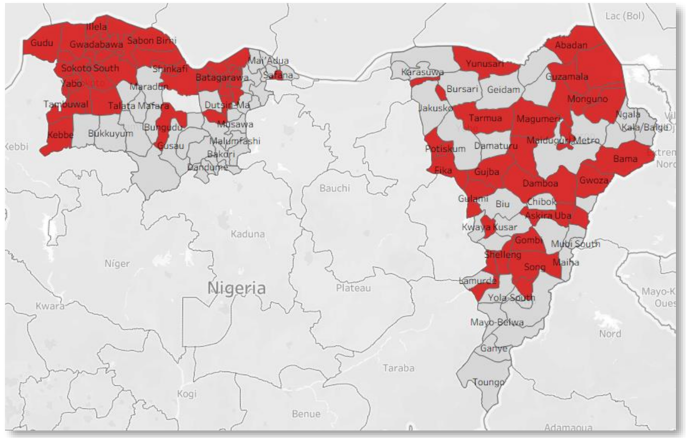
- Identifies and tracks key drivers of food insecurity to estimate current conditions between CH assessments

Performance and validation

- Validated against CH observation points with an error of 5–6%, confirming strong alignment



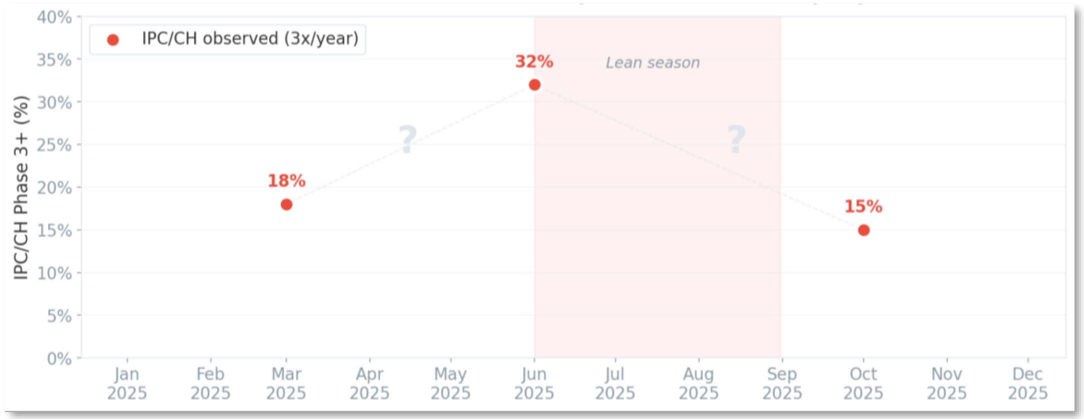
Blue line: nowcast - **Dots:** CH assessment



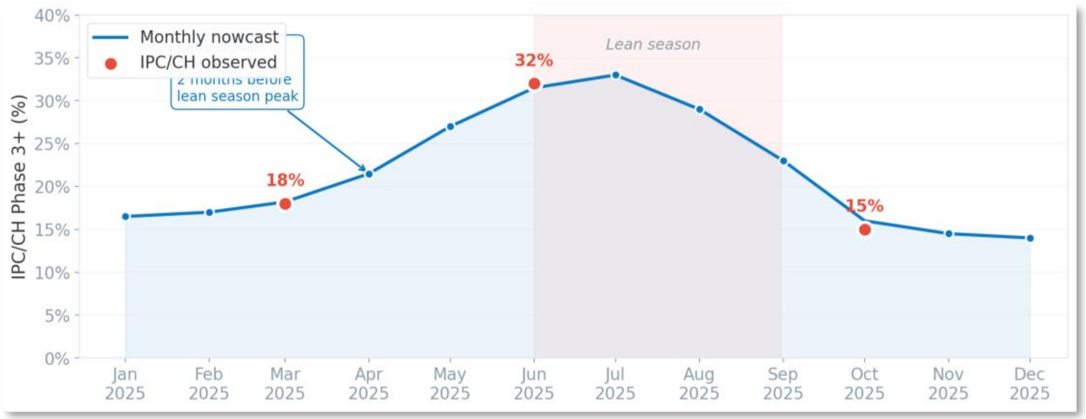
LGAs that would have triggered a warning in the latest period based on nowcasting estimates

Tracking Food Insecurity Between Assessment

CH assessments published 2-3 times per year

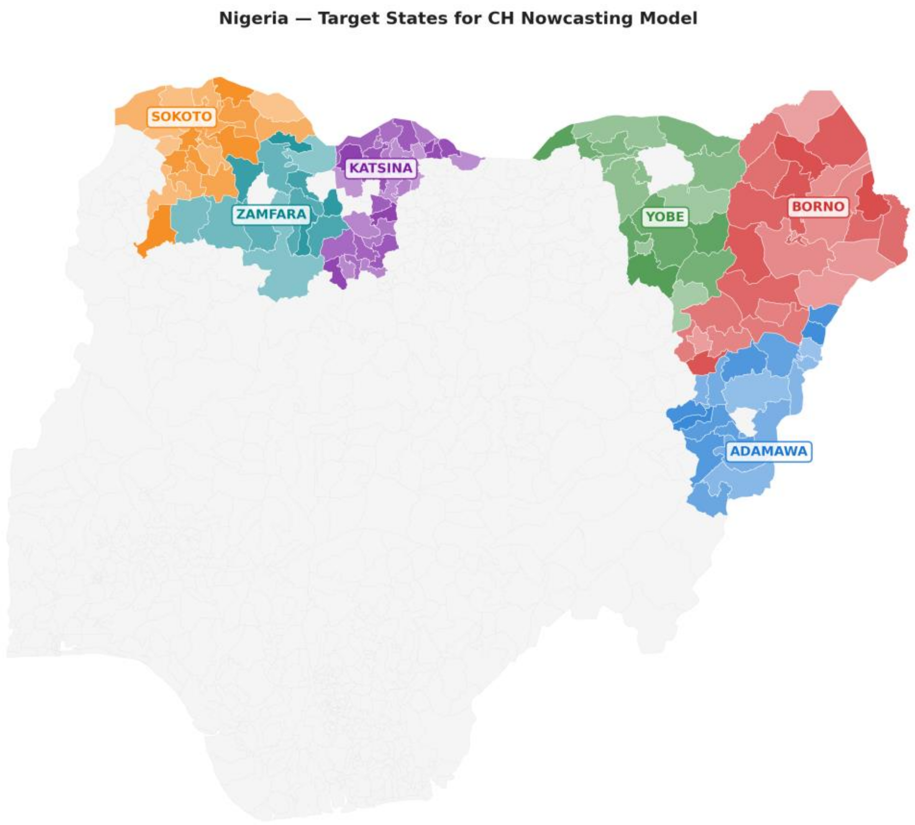


Nowcasting: continuous monthly monitoring



The model provides continuous, explainable monthly food insecurity estimates by filling the gaps between CH assessments and showing how each predictor contributes to the nowcast.

What the model covers



Outcome

Predict % of population in CH Phase 3+ at LGA level

Where

Six high-risk northern states - Adamawa, Borno, Yobe, Katsina, Sokoto, Zamfara

How

Continuous monitoring of drivers to detect changes early - conflict, markets, and climate stressors

Country Level

Date	IPC/CH 3+ (%)	Estimation	Error
2025-03	19.80	19.58	-0.22
2025-06	25.07	18.82	-6.26
2025-10	21.82	18.71	-3.11

WFP
World Food Programme
SAVING LIVES
CHANGING LIVES

How Well the Nowcasting Model Performs

5.3

Model Error

The model's predictions are, on average, 5.3 pp away from the actual CH values observed at LGA level.

6.4

Baseline error

Assuming conditions haven't changed since the last assessment results in a larger error than the model.

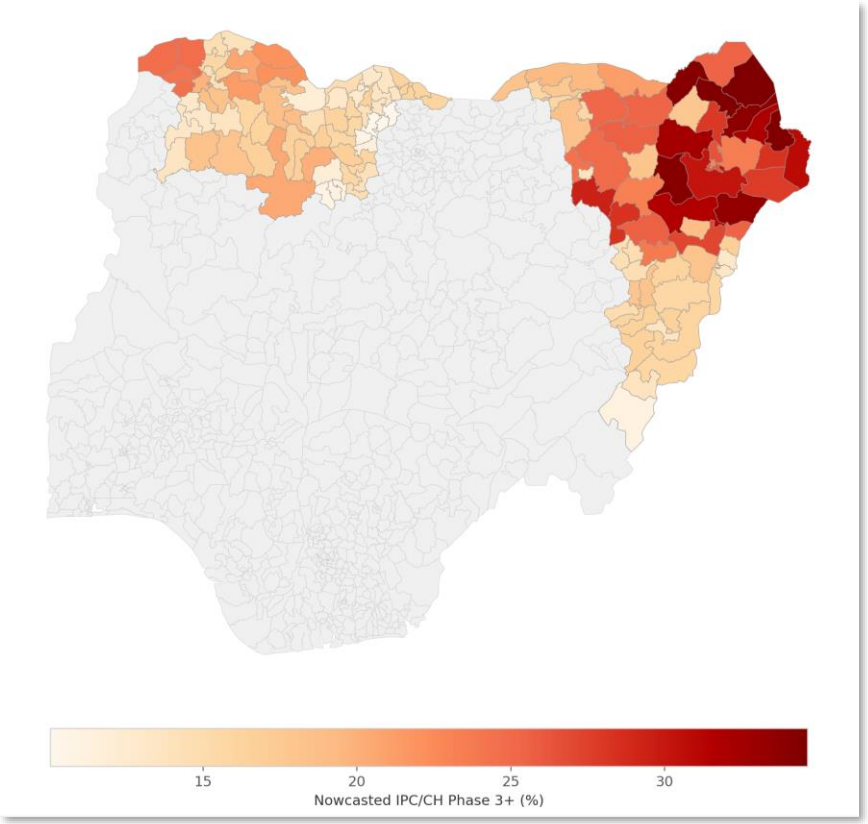
7.1

Seasonal error

Assuming this year looks like the same season last year performs even worse, with the highest error.

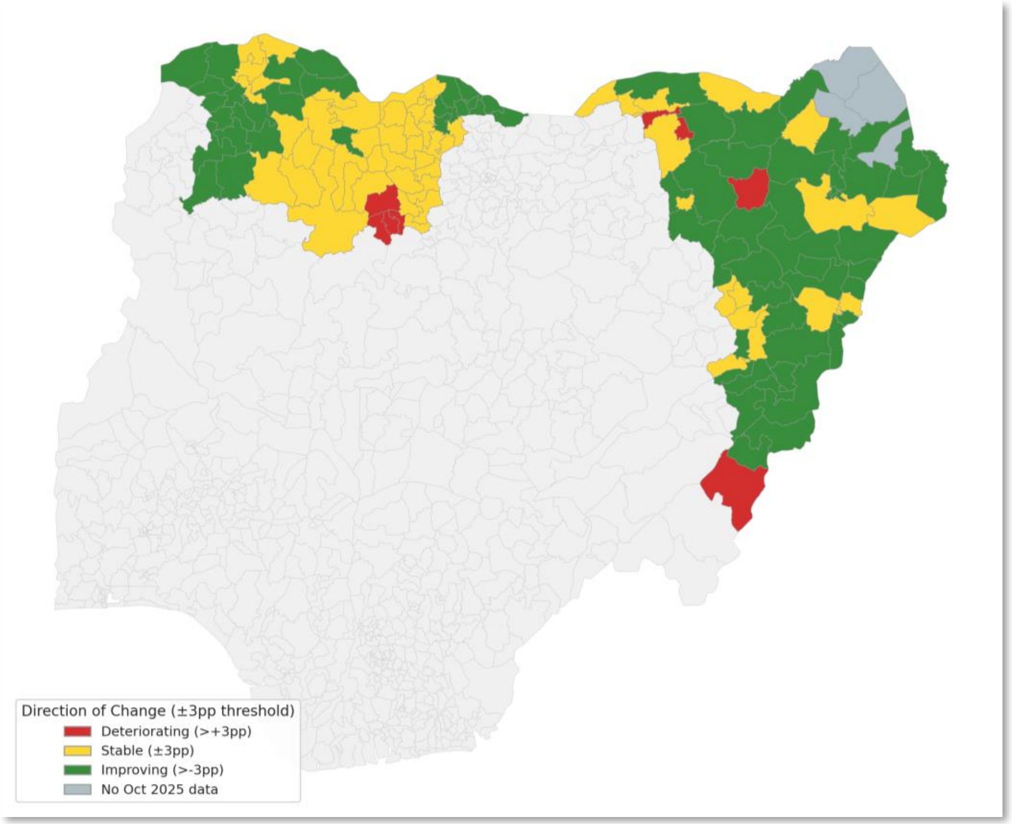
March 2026 Nowcast by LGAs

CH 3+ prevalence – March 2026



Borno LGAs remain the most food insecure in March 2026, with several exceeding 30% Phase 3+. Northwestern states show moderate levels.

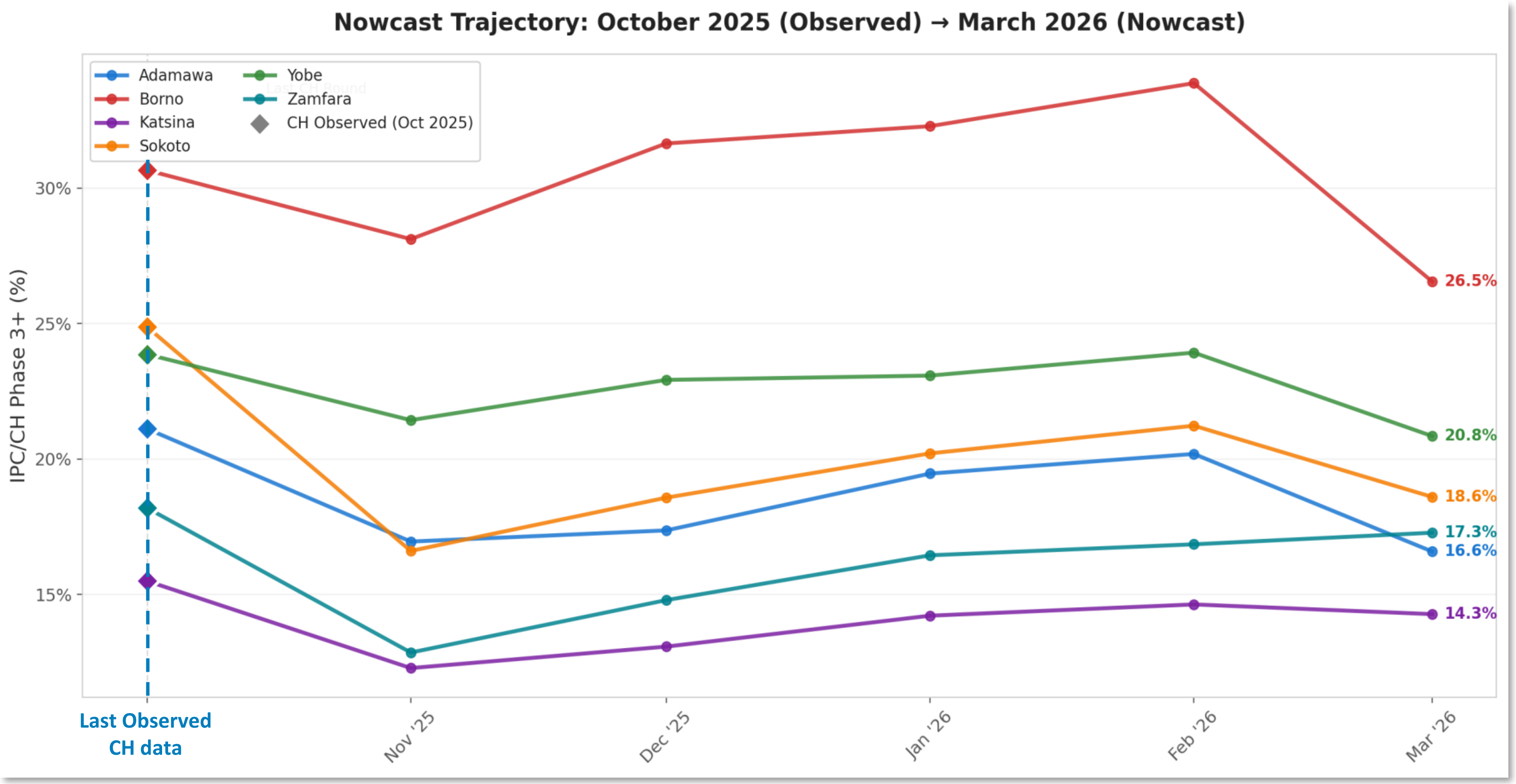
CH 3+ prevalence – Change since last observed CH



Most LGAs show seasonal improvement from October's peak.
7 LGAs continue deteriorating despite the post-harvest period, concentrated in Katsina and Borno.
Main drivers: affordability and conflict

Monthly Nowcast trends by State

How food security conditions change month by month, using the nowcasting model to fill the gap after the last available assessment



Micronutrient Inadequate Intake

model: Cross Country Prediction / Nowcasts
Granularity: adm1/adm2

HungerMap LIVE

Dietary Micronutrient Inadequacy

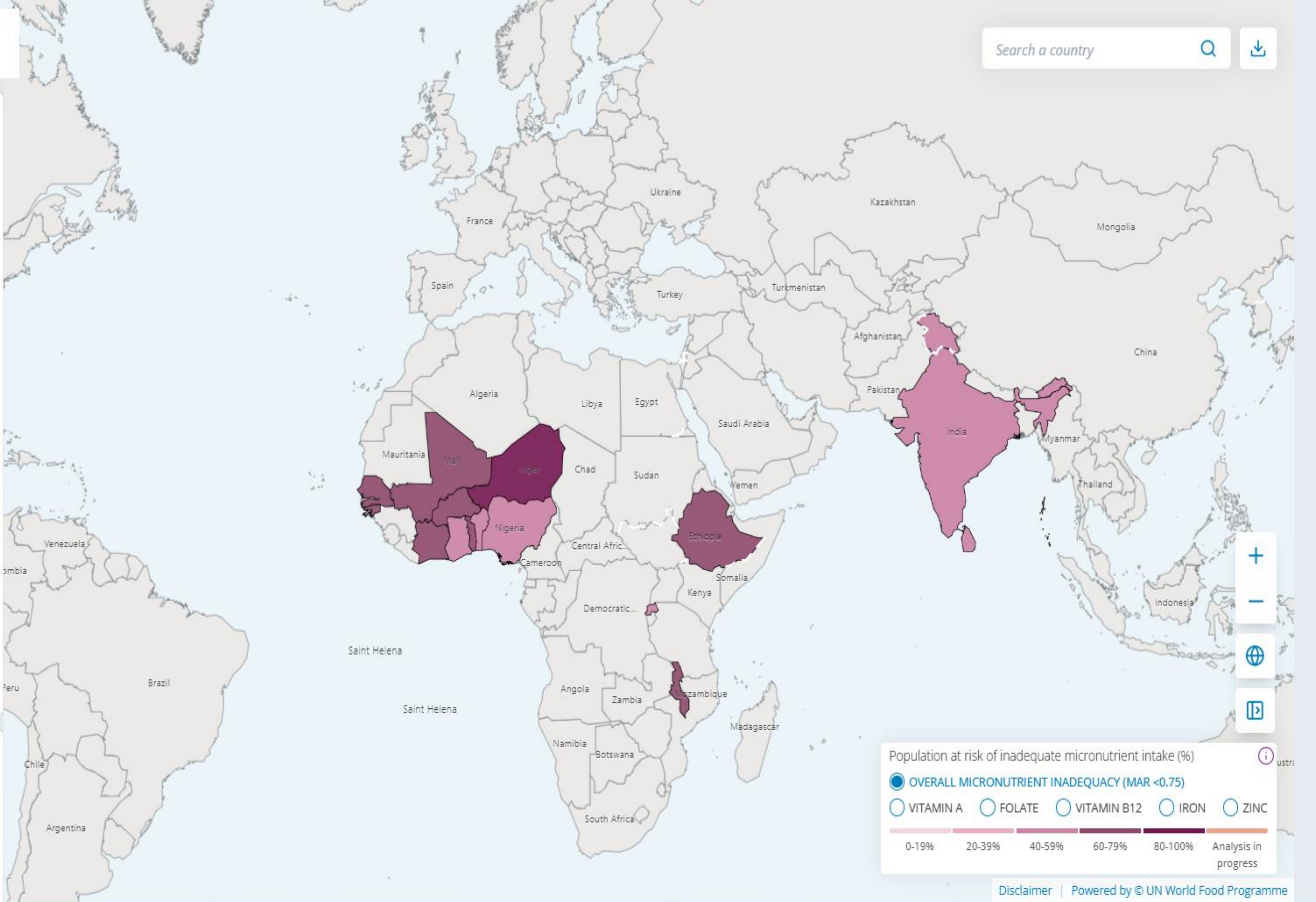
Top countries by risk of inadequate micronutrient intake (Overall Mean Adequacy Ratio (MAR < 0.75))

NIGER	88.0%
MALI	79.0%
GUINEA-BISSAU	73.0%
ETHIOPIA	72.0%
BURKINA FASO	70.0%
CÔTE D'IVOIRE	68.0%
TOGO	68.0%
SENEGAL	62.0%
MALAWI	61.0%
BENIN	58.0%

SEE THE FULL DATA

Range of households with inadequate micronutrient intake (%) across countries modelled

FOLATE 5% - 88%



Disclaimer | Powered by © UN World Food Programme

THE PROBLEM

The Micronutrient Data Gap

The Gap

Micronutrient inadequacy is widespread but poorly mapped at subnational level in most low-income countries.

The Barrier

Gold-standard measurement requires quantitative 7D dietary recall — expensive, slow, and rarely conducted.

The Untapped Resource

WFP routinely collects food security surveys in dozens of countries — but these aren't designed for micronutrient assessment.

Core question: Can routine food security data predict micronutrient inadequacy — and does that signal transfer across countries?

DATA & FEATURES

What Goes Into the Model

Features

All available in routine WFP food security surveys

FCS-N Food Groups
16 binary consumed/not-consumed groups incl. vitamin A rich, protein sub-groups

Crop & Livestock Production
Binary indicators for maize, millet, sorghum, rice, cassava, teff, cattle, poultry, etc.

Expenditure Shares
Continuous proportions by food group (staples, pulses, dairy, protein, veg, fruit, fat, sugar) and source (own-production – market)

Demographics & Wealth
Education level, IWI (cross-country comparable), dependency ratio

Targets

Binary inadequacy from 7 Day recall

- Folate
- Zinc
- Iron
- Vitamin B12
- Vitamin A
- Overall MAR

Design principle: every feature is mechanistically linked to nutrition and measured on comparable scales across countries → ensures transferability.

Example of Machine Learning models to predict risk of inadequate micronutrient intake

MIMI
MODELLING AND MAPPING INADEQUATE MICRONUTRIENT INTAKE

HCES and other data sources

Predictive variables:

- Food group diversity
- Housing
- Education
- Climate
- Food price shocks

Target to be predicted:

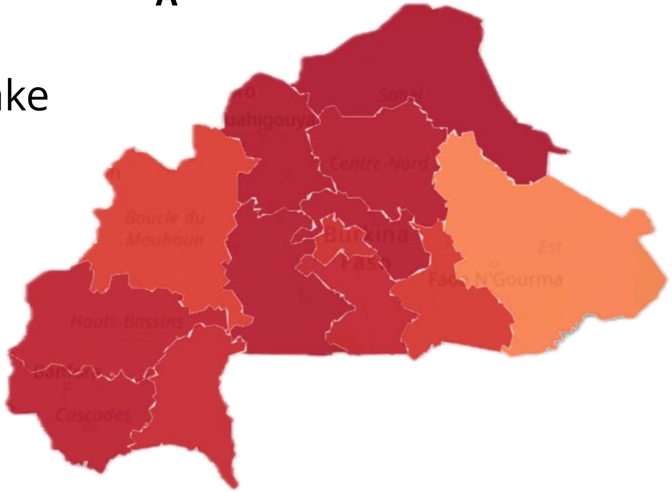
- Estimated risk of inadequate micronutrient intake

Machine learning model trained

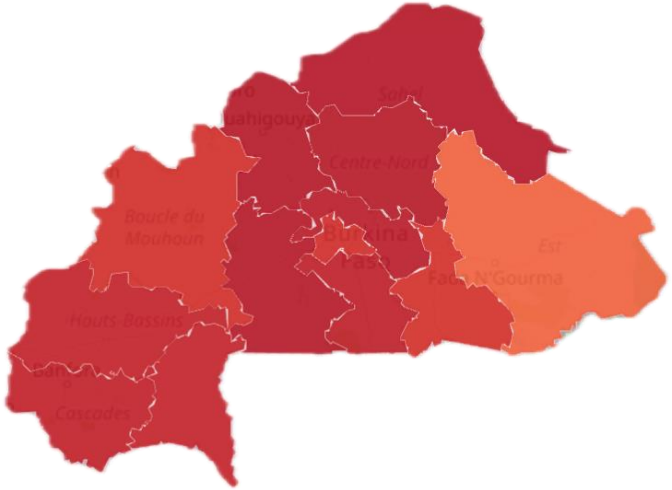
Actual input data from routine Food Security and climate monitoring

Burkina Faso

Predicted Vitamin A

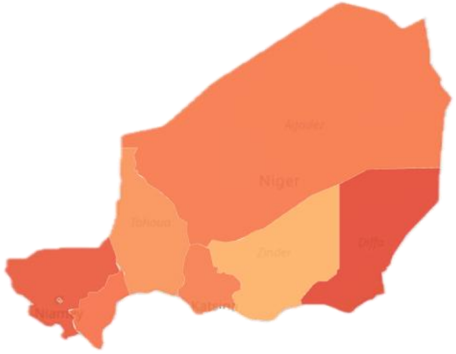


Actual Vitamin A

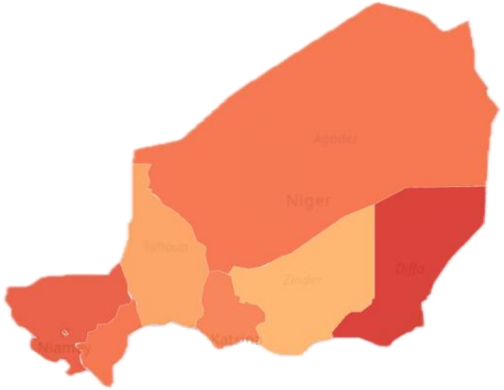


Niger

Predicted Zinc



Actual Zinc



Source: Voukelatou et al, 2026

Methodological approach

scientific reports

Article | Open access | Published: 10 January 2026

Predicting risk of inadequate micronutrient intake with transferable machine learning models

Vasiliki Voukelatou, Kevin Tang, Ilaria Lauzana, Manita Jangid, Giulia Martini, Saskia de Pee, Frances Knight & Duccio Piovani

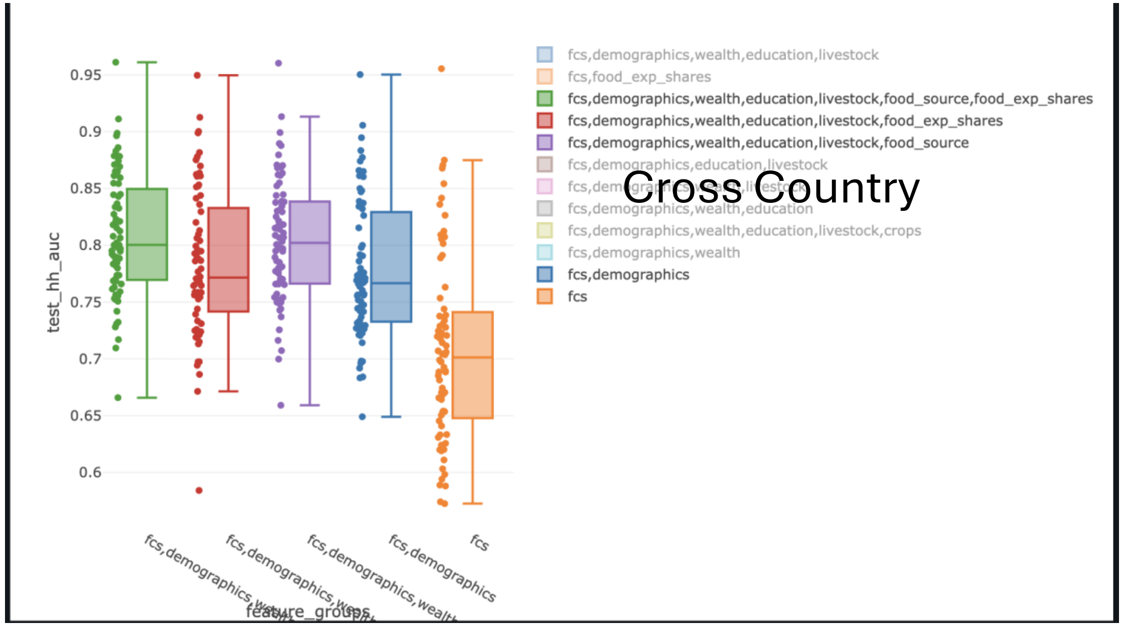
Scientific Reports **16**, Article number: 4104 (2026) | Cite this article

Within Country

Cross Country

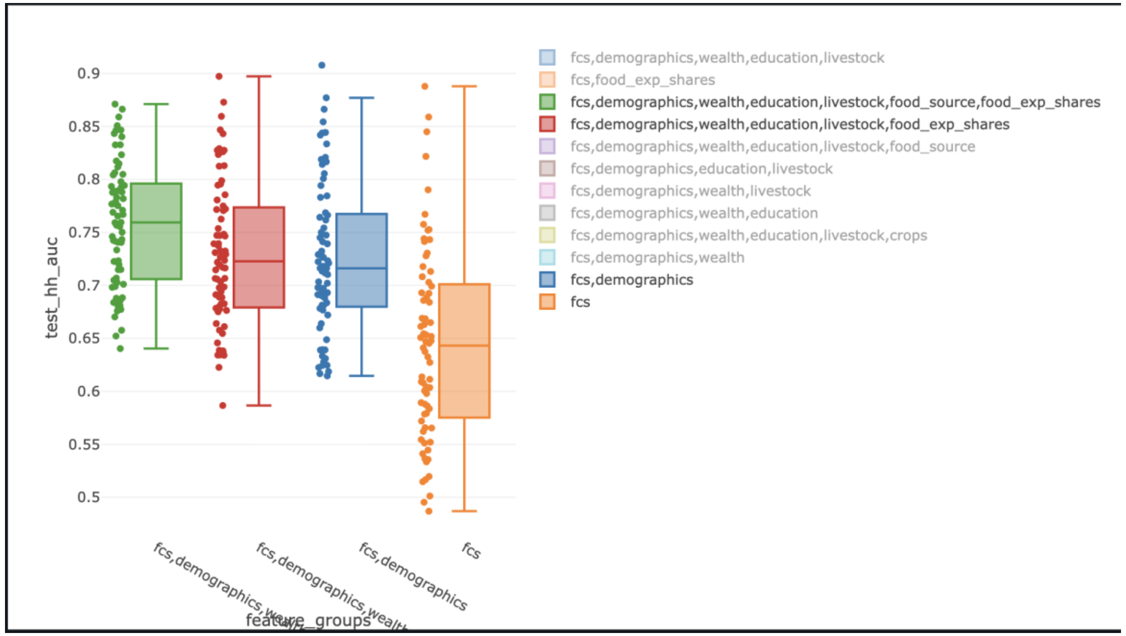
Median AUC = 0.8

Food Source does a lot of heavy lifting



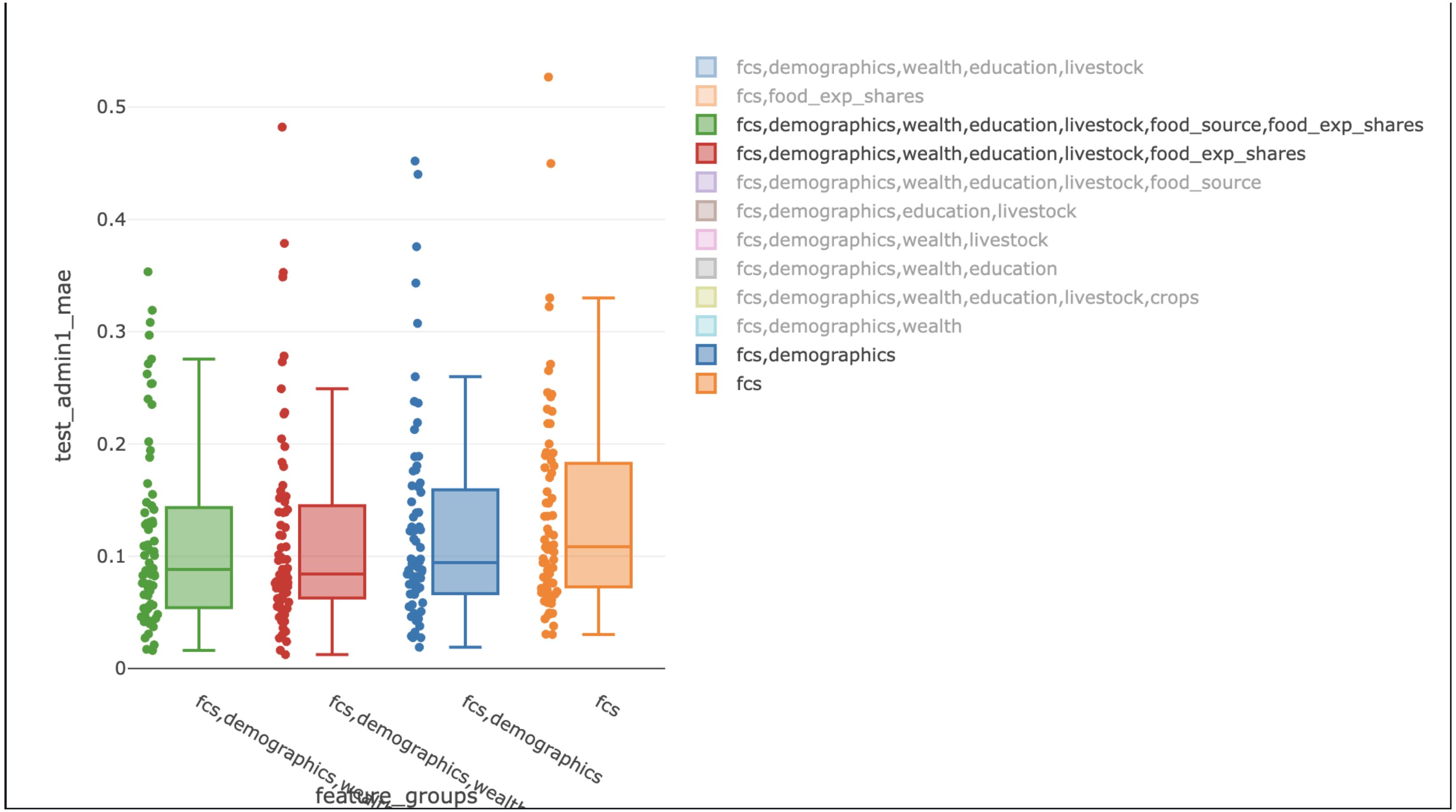
Median AUC = 0.75

Remarkably close to within country performance!

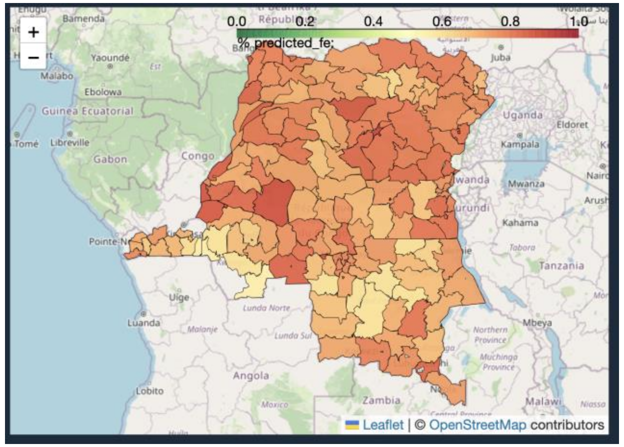


Admin Level Prevalence is another story...

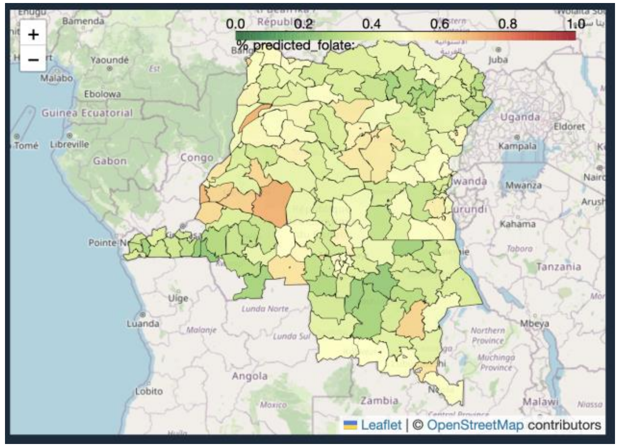
Median MAE from 4.5% → 8.5%



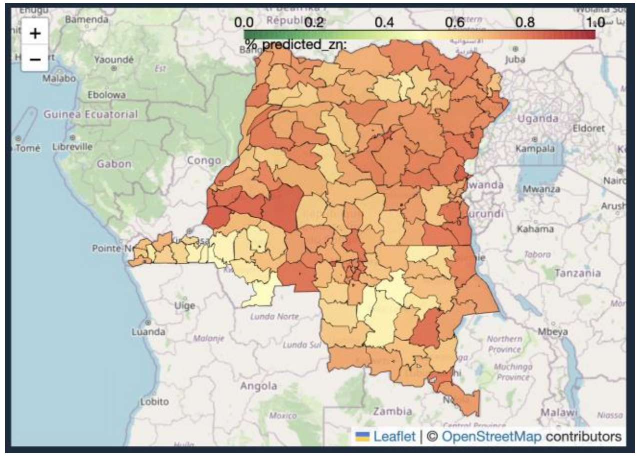
Iron



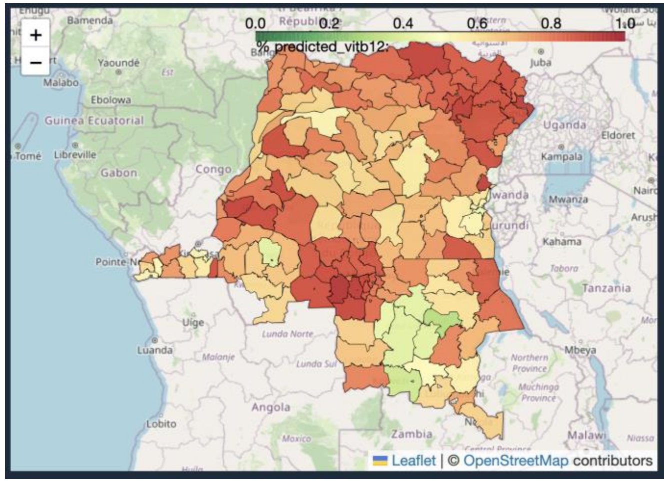
Folate



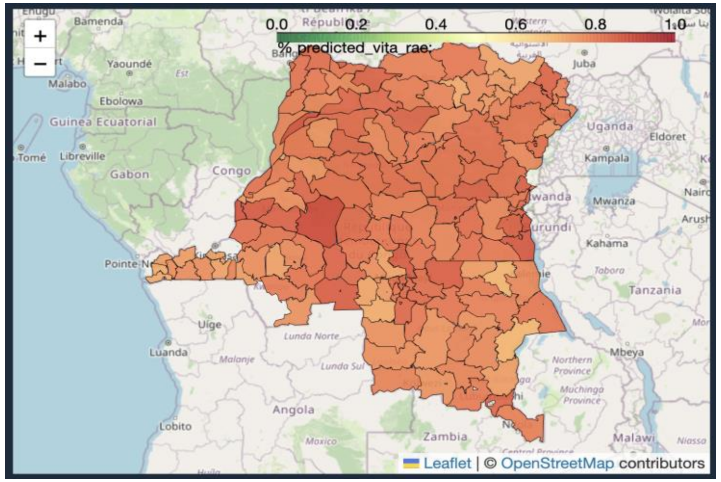
Zinc



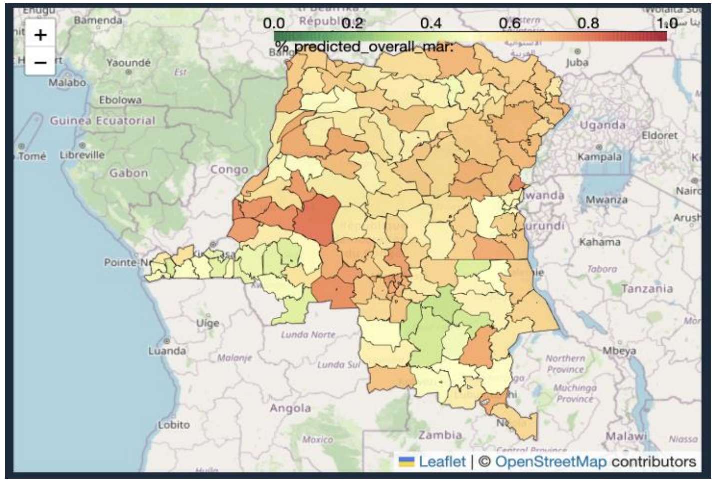
Vitamin B12



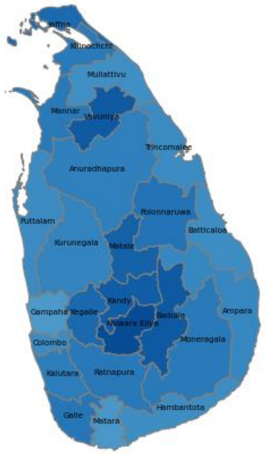
Vitamin A

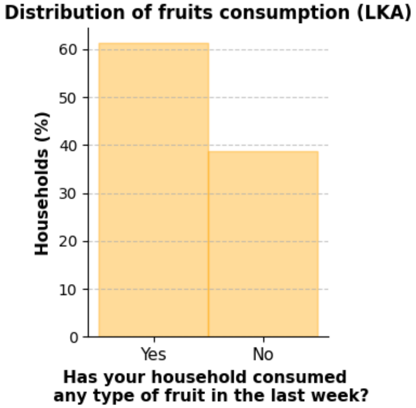


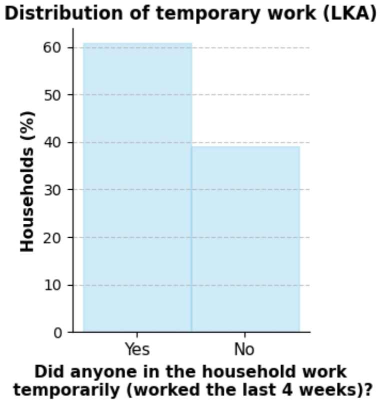
Overall MAR



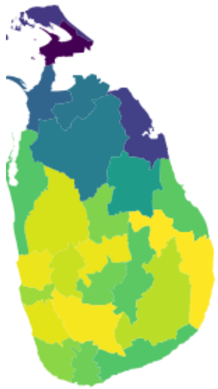
Risk of inadequate overall intake in 2019







Median rainfall between 2017 and 2019, at 2nd level administrative unit resolution.



Target

Available **only** in dietary survey data

- Household-level survey data
- Quantitative (g per week) household consumption of specific food items (e.g., 1. yogurt, 2. wheat flour, etc)

Machine-learning

Features (Variables)

Food group diversity

- Household level survey data
- Binary (yes/no per week) household consumption of 9 food groups (e.g., 1. milk and other dairy products, 2. cereal, grains, roots and tubers, etc)

Socio-economic status

- Household level survey data
- Binary or categorical education-, labour-, housing conditions-, and assets ownership-related household level data

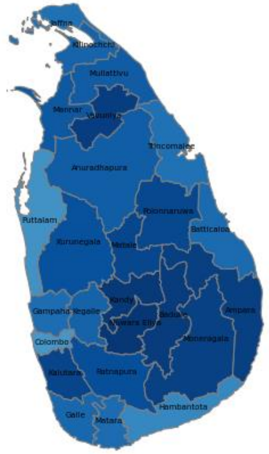
Climate

- Satellite data
- Rainfall, rainfall anomaly, and Normalized Difference Vegetation Index (NDVI) time-series

Prices

- Prices of food commodities
- Data collected from HARTI (Hector Kobbekaduwa Agrarian Research and Training Institute)

Available both in dietary survey data and VAM survey data.



Risk of inadequate overall intake in 2024

Early Warning of Food Security AFG

modeling: Nowcast
Granularity: adm1/adm2

Early Warning System > Afghanistan – admin2 & 3

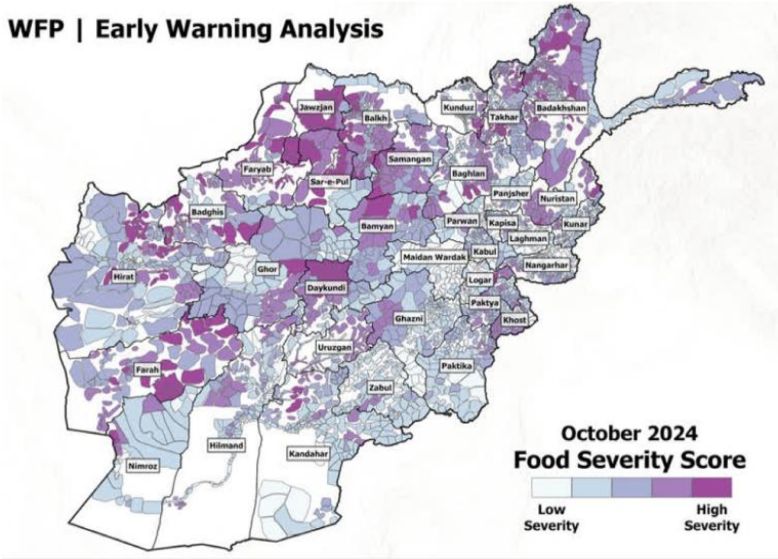
Overview

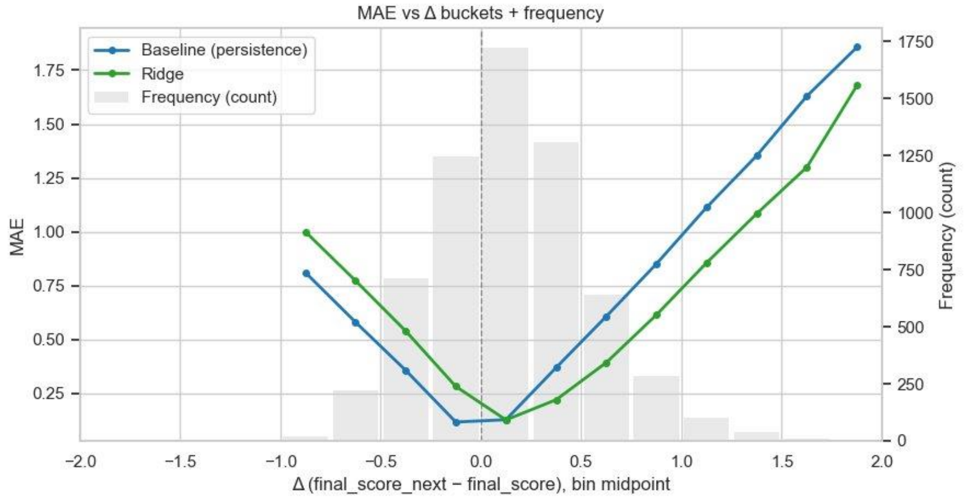
- Building on the Early Warning model developed by the country office.
- End-to-end early-warning system integrates multiple data sources into a risk index linked to programmatic action.
- Data is organized in 5 pillars: access, availability, utilization, stability

Model Overview

- Adding a forward-looking component to the existing early warning model, forecasting future values of the risk index.
- Automation of the early warning systems embedding in HQ infrastructure

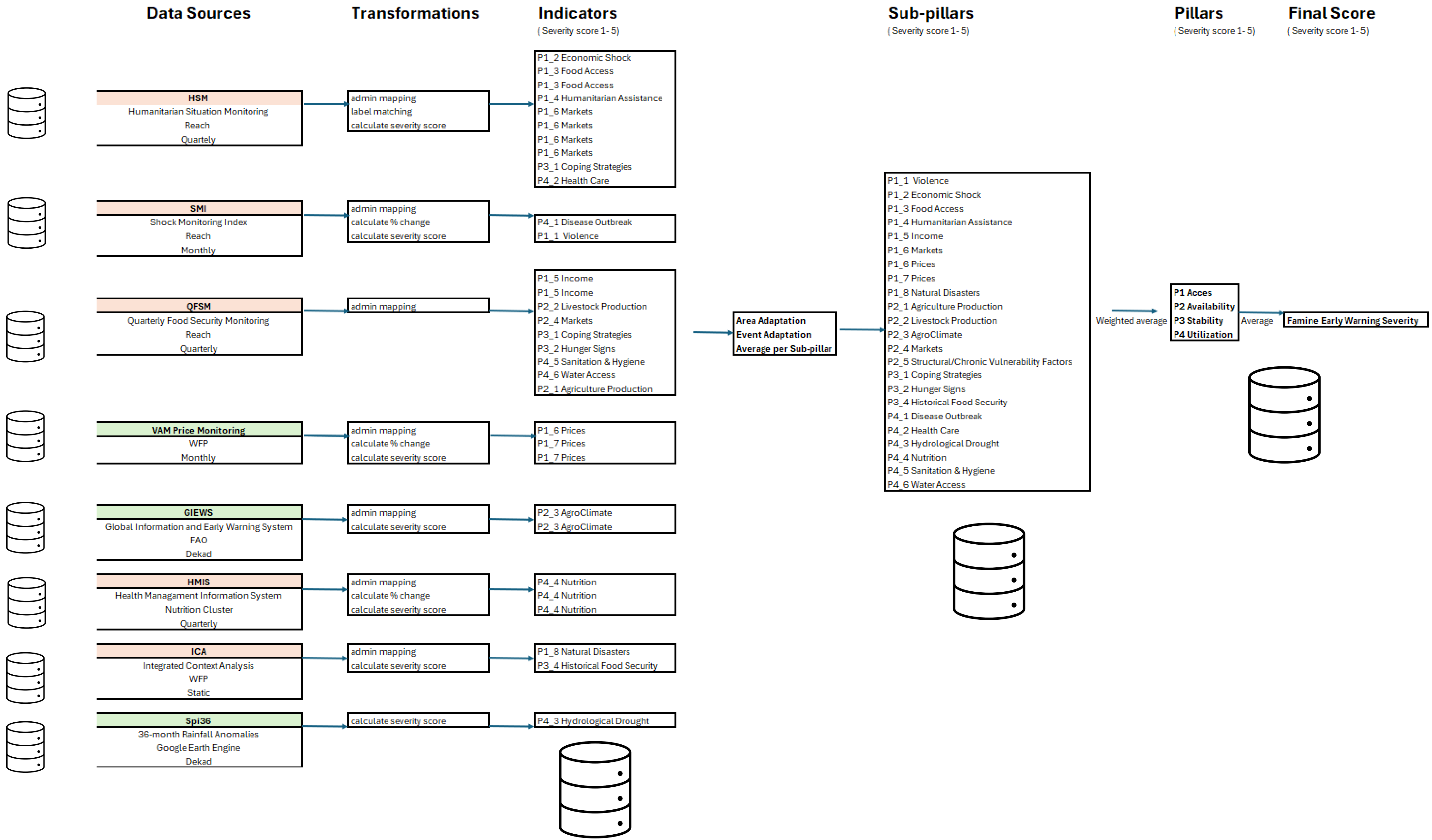
Food Security Index - Afghanistan





Analytical Framework

Pillar	Sub-Category	Indicator	Data Source	Frequency
P1 Access	P1_3 Food Access	% of KIs reporting households are not able to access enough food to meet daily needs within 30 days of data collection, per proportion of households	REACH	Quarterly
P1 Access	P1_6 Markets	% of settlements where KIs reporting the three most common challenges for women regarding access to markets	REACH	Quarterly
P1 Access	P1_6 Prices	Increase of minimum food basket price compared with previous month	VAM Market Bulletin & Joint Market Monitoring Initiative	Monthly
P2 Availability	P2_1 Agriculture Production	% of settlements by proportion of households having experienced a decrease in harvesting of HALF or More than half of their production in the last harvest period (among settlement reporting agriculture as a primary or secondary income source)	REACH	Quarterly
P2 Availability	P2_2 Livestock Production	% of settlements by proportion of households having experienced a large decrease in the number of livestock owned in the past 3 months (among settlement reporting livestock as a primary or secondary income source)	REACH	Quarterly
P2 Availability	P2_3 AgroClimate	Crops: Weighted Mean Vegitation Health Index	FAO	10 Days
P2 Availability	P2_3 AgroClimate	Pastoral: Weighted Mean Vegitation Health Index	FAO	10 Days
P3 Stability	P3_1 Coping Strategies	% of settlements with households involuntarily moving due to lack of food	REACH	Quarterly
P3 Stability	P3_1 Coping Strategies	Approximately what proportion of households in the settlement of interest have school-aged children engaging in employment outside of their homes? (% of settlements where all or many households have school-aged children engaging in employment outside their homes)	REACH	Quarterly
P3 Stability	P3_2 Hunger Signs	% of settlements where the KIs reporting on the perceived level of hunger among the households	REACH	Quarterly
P4 Utilization	P4_1 Disease Outbreak	Disease Outbreak	SMI	Montlhy
P4 Utilization	P4_3 Hydrological Drought	Severity of hydrological drought in settlement	WFP	10 Days
P4 Utilization	P4_4 Nutrition	SAM admission for the last quarter against the same quarter of the last year	Nutrition Cluster	Quarterly


Data Sources
Transformations
Indicators
(Severity score 1- 5)
Sub-pillars
(Severity score 1- 5)
Pillars
(Severity score 1- 5)
Final Score
(Severity score 1- 5)
HSM
Humanitarian Situation Monitoring
Reach
Quartely
admin mapping
label matching
calculate severity score
P1_2 Economic Shock
P1_3 Food Access
P1_3 Food Access
P1_4 Humanitarian Assistance
P1_6 Markets
P1_6 Markets
P1_6 Markets
P1_6 Markets
P3_1 Coping Strategies
P4_2 Health Care
SMI
Shock Monitoring Index
Reach
Monthly
admin mapping
calculate % change
calculate severity score
P4_1 Disease Outbreak
P1_1 Violence
QFSM
Quarterly Food Security Monitoring
Reach
Quarterly
admin mapping
P1_5 Income
P1_5 Income
P2_2 Livestock Production
P2_4 Markets
P3_1 Coping Strategies
P3_2 Hunger Signs
P4_5 Sanitation & Hygiene
P4_6 Water Access
P2_1 Agriculture Production
VAM Price Monitoring
WFP
Monthly
admin mapping
calculate % change
calculate severity score
P1_6 Prices
P1_7 Prices
P1_7 Prices
GIEWS
Global Information and Early Warning System
FAO
Dekad
admin mapping
calculate severity score
P2_3 AgroClimate
P2_3 AgroClimate
HMIS
Health Managament Information System
Nutrition Cluster
Quarterly
admin mapping
calculate % change
calculate severity score
P4_4 Nutrition
P4_4 Nutrition
P4_4 Nutrition
ICA
Integrated Context Analysis
WFP
Static
admin mapping
calculate severity score
P1_8 Natural Disasters
P3_4 Historical Food Security
Spi36
36-month Rainfall Anomalies
Google Earth Engine
Dekad
calculate severity score
P4_3 Hydrological Drought
Area Adaptation
Event Adaptation
Average per Sub-pillar
P1_1 Violence
P1_2 Economic Shock
P1_3 Food Access
P1_4 Humanitarian Assistance
P1_5 Income
P1_6 Markets
P1_6 Prices
P1_7 Prices
P1_8 Natural Disasters
P2_1 Agriculture Production
P2_2 Livestock Production
P2_3 AgroClimate
P2_4 Markets
P2_5 Structural/Chronic Vulnerability Factors
P3_1 Coping Strategies
P3_2 Hunger Signs
P3_4 Historical Food Security
P4_1 Disease Outbreak
P4_2 Health Care
P4_3 Hydrological Drought
P4_4 Nutrition
P4_5 Sanitation & Hygiene
P4_6 Water Access
Weighted average
P1 Acces
P2 Availability
P3 Stability
P4 Utilization
Average
Famine Early Warning Severity

1. Sub-District food security index

DIMENSIONS OF FOOD SECURITY

Food Access

Household incomes, market prices, women's social access to markets, shocks (economic, conflict & natural hazards), humanitarian assistance

Food Availability

Market functionality, agricultural production, livestock production, drought, water and crop growth

Food Utilization

Access to safe water, sanitation and hygiene, acute malnutrition estimates (SAM and MAM admissions trends)

Stability

Livelihood coping strategies, distress migration, household debt

Research and Preparation

1. Assessing the Data Landscape
2. Filling in the Data Gaps
3. Understanding the most significant drivers of food security according to the country
4. Continuous learning and adapting

Research Gaps

- Historical analysis of droughts
- Gozar mapping for urban centers
- Crop mapping
- Nomadic pastoralists movement and locations
- Non-farm income opportunities

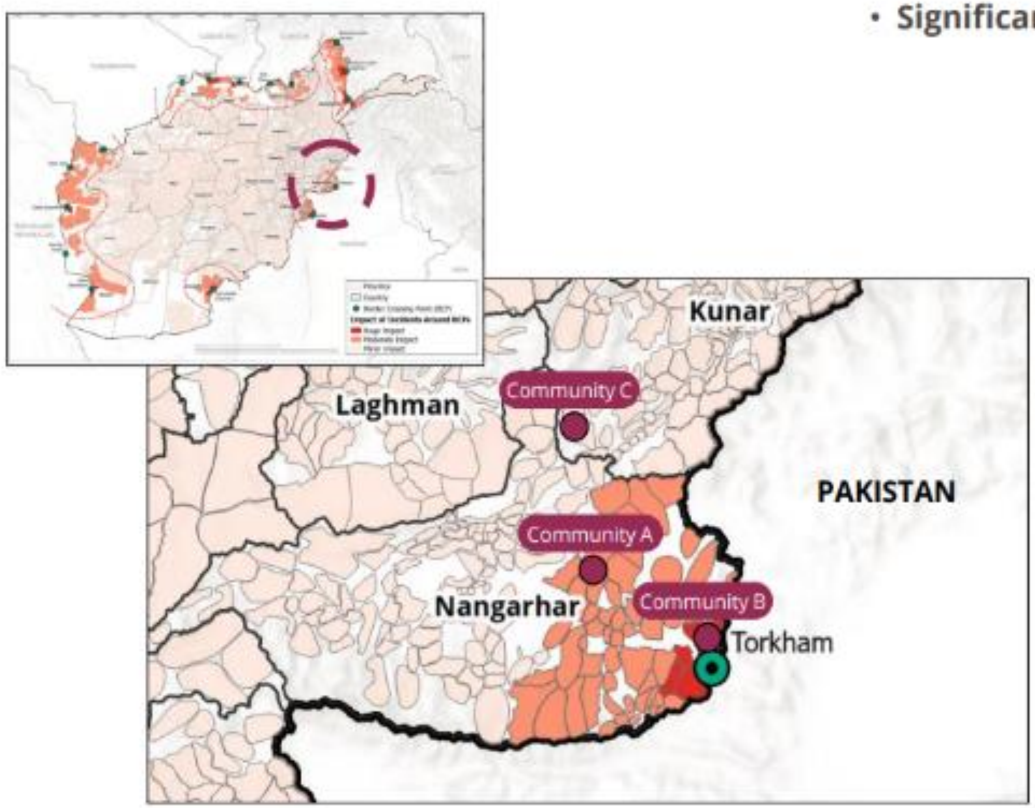
FOOD SEVERITY SCORE

Area Adaptation

Event-based Trigger

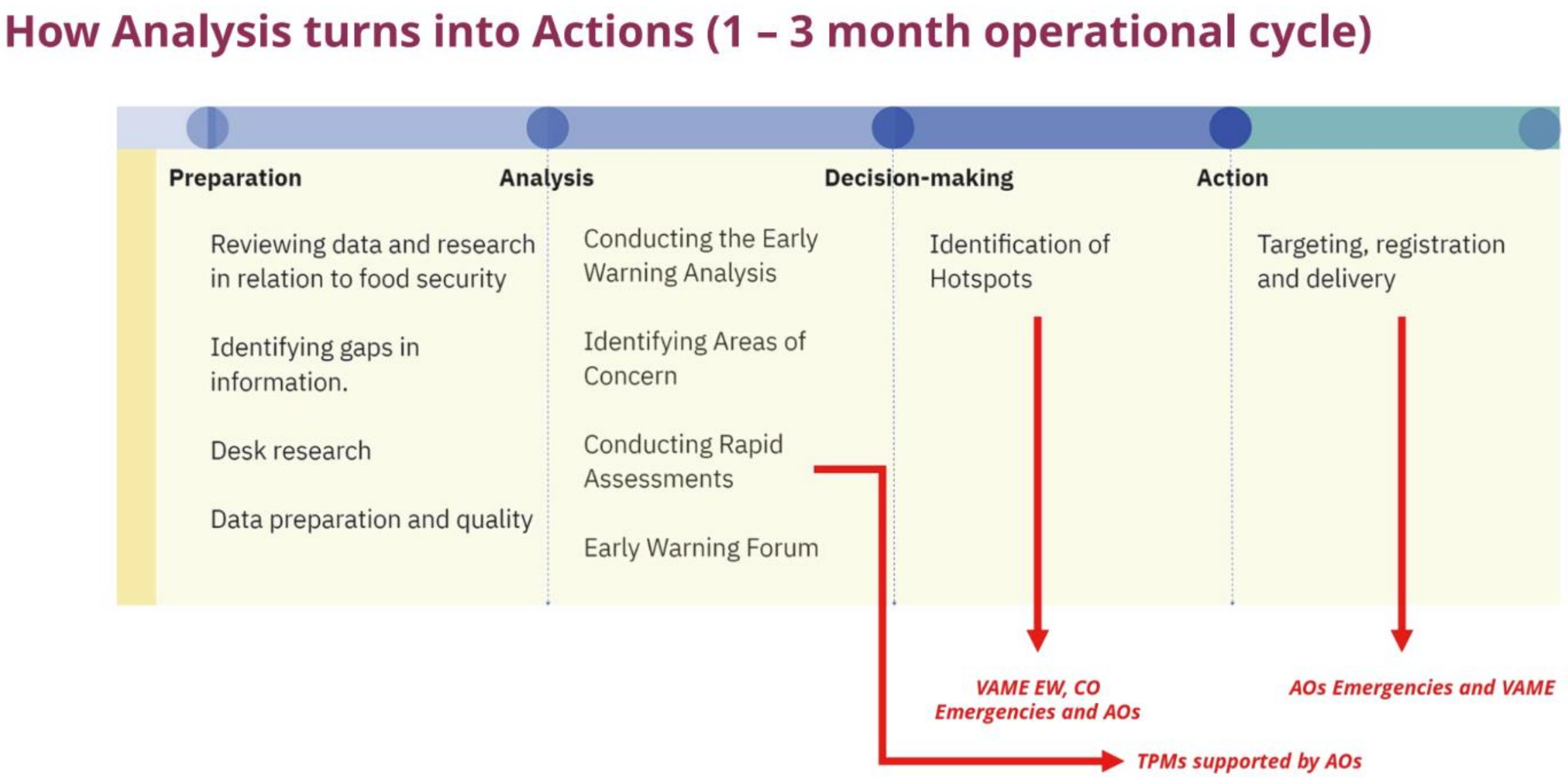
Scenario

- **3 different communities near "Torkham" border crossing points** with Pakistan.
- Areas are located **within different distances** from the border crossing point.
- **Same severity** for access to health facilities.
- **Significant returnees Influx is anticipated** within analysis period (**Shock**)



	Absolute Severity Score		Adaptation Factor		Final Severity Score
Community A	3	x	1	=	3
Community B	3	x	1.5	=	4.5
Community C	3	x	0.5	=	1.5

Analytical Framework



Conflict Forecasting modeling: Ensemble Modeling

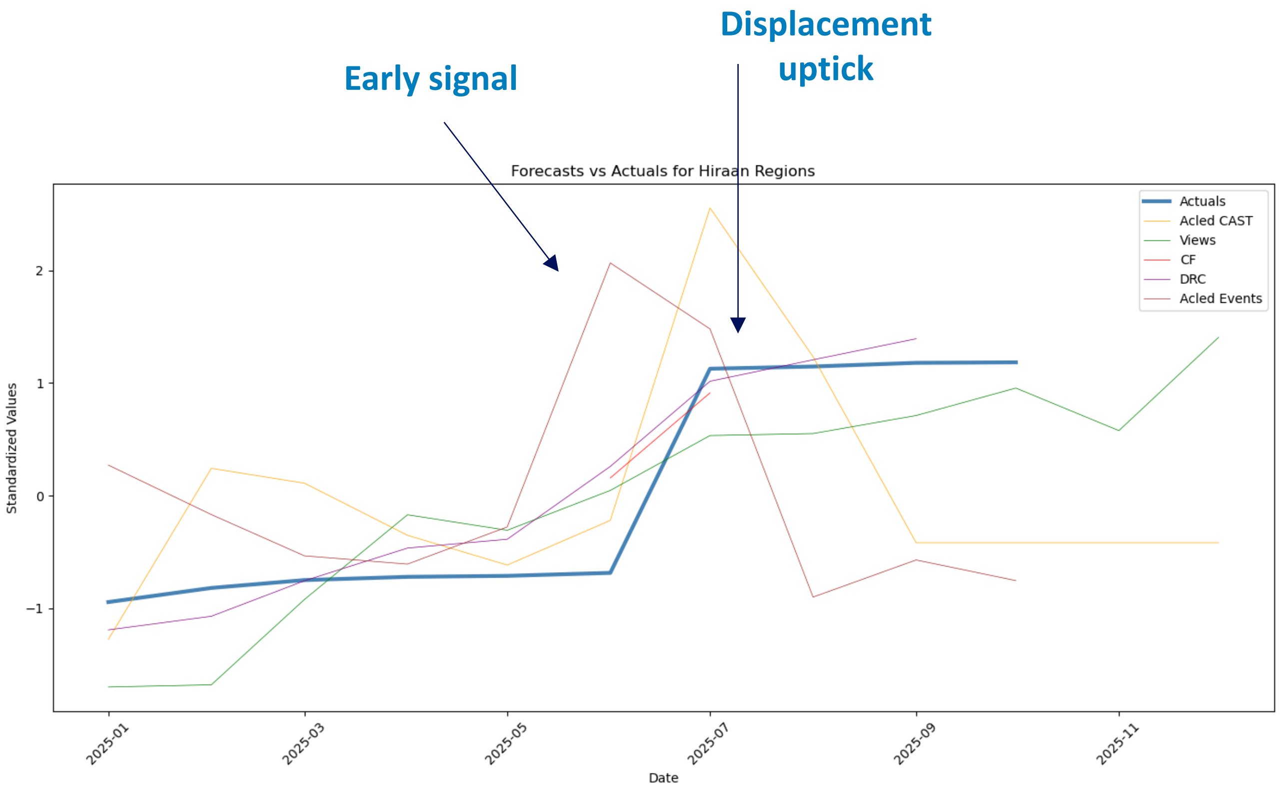
Conflict Forecast Ensemble Model Somalia – admin2

Model Overview

- Develop an ensemble forecasting model that integrates conflict and IDP prediction tools developed by established institutions.
- Inputs: Conflict and displacement forecasts (**Acled CAST**, **VIEWS**, **ConflictForecast**, **Danish Refugee Council)**
- The goal is to provide a tool that delivers signals to flag potential deterioration in food security conditions at the Admin2 level.

Early Results (phase 1):

- **correlations with displacement upticks provided by UNHCR (see figure)**
- **ACLED CAST, VIEWS, and DRC** show the highest correlation with actual time series.
- On average, these tools can forecast upticks **2 to 4 months in advance**.



Phase 1: Analysis on the relationship between conflict forecast upticks and upticks in displacement.
Phase 2: relation with Food Insecurity (IPC)
Phase 3: Ensemble model (TBD)
In the graph: July 25 in Hiraan, where 46,000 new displacements were recorded

Conflict/Displacement Forecasting Tools

Conflict

Acled CAST	Predicts number of events at admin1 – 6 months
VIEWS	Predicts number of fatalities and conflict probability at subnational level - 36 months
EconAI/ Conflict forecast	Predicts probability of conflict/violence and n. fatalities at subnational level – 3/12 months

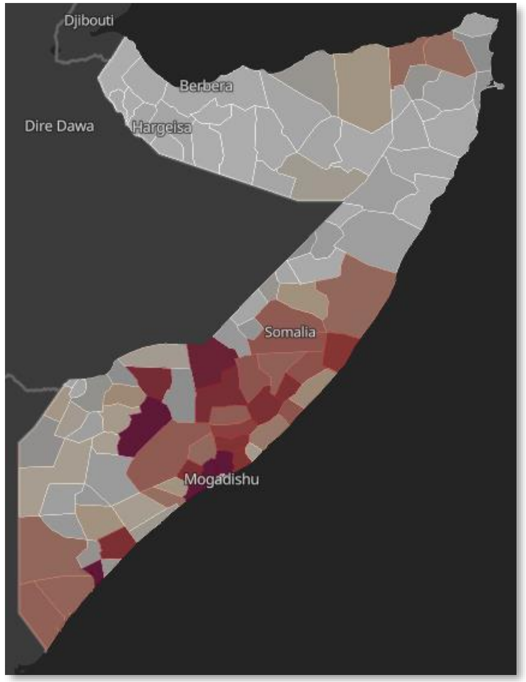
Internal Displacement

DRC - PREDICT	Predicts number of total IDPs at district level - 3 months
EconAI	Predicts escalations where the number of IDPs increases from below to above the 1,000 threshold – country level - monthly

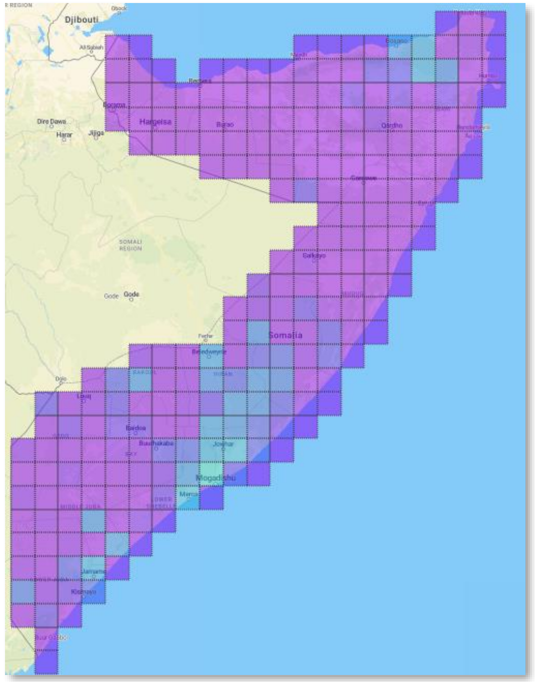
Forecasting tools differ in their methodologies and characteristics, which directly influence the type of predictions they generate, as well as their temporal and spatial resolution

Challenge: Combine highly different outputs into a single ensemble model

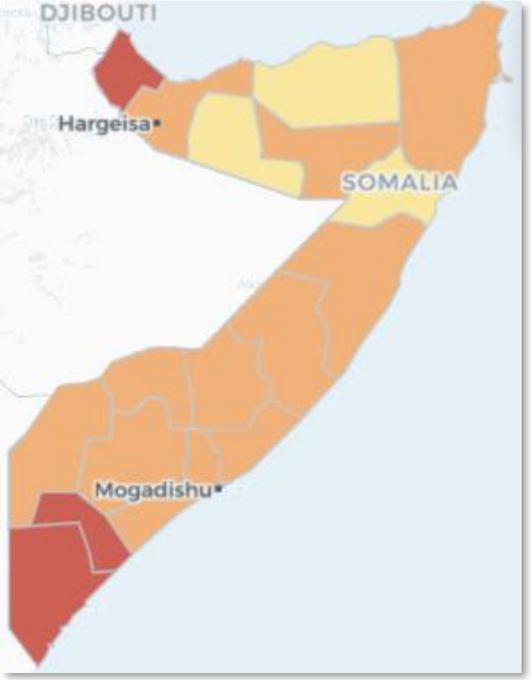
EconAI - ConflictForecast



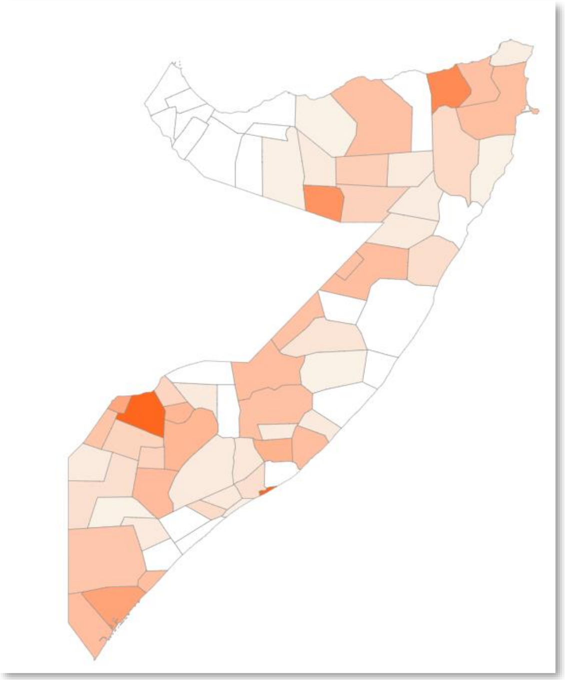
Views



Acled CAST



DRC – displacement forecast



HML Demo

World Food Programme

SAVING LIVES
CHANGING LIVES

Identified Best Practices (again!)

- **Start from decisions, not models**: Clearly define the operational or strategic decisions the early-warning system is meant to inform, and design models to support those decisions rather than prediction for its own sake.

- **Embed domain expertise from the outset**: Food security and nutrition experts must be involved from problem formulation through interpretation, ensuring models reflect livelihoods, seasonality, shocks, and operational realities.

- **Explicitly map needs to decision pathways**: Identify the specific needs being targeted and the decision-making processes to be influenced, including who acts, at what geographic level, and on what time horizon.

- **Be realistic about data constraints**: Align modelling ambition with what available data can credibly support in terms of quality, coverage, frequency of updates, and timeliness, prioritizing robustness over false precision.

- **Use models to structure risk, not replace judgement**: Modelling outputs should support transparent, risk-based discussions and human-in-the-loop decisions, rather than automate responses in complex food security contexts.

Thank you !

WFP

World Food Programme

SAVING LIVES

CHANGING LIVES